Abstract

Automated segmentation supports post-treatment glioma assessment, yet limited expert capacity constrains quality control. We developed a multimodal triage system combining an MRI-derived segmentation-risk score with time since radiotherapy, IDH status and MGMT promoter methylation to prioritise segmentations for review. Following retrospective development in 50 patients, the locked system was evaluated prospectively in silent mode in 80 consecutive patients. At a prespecified 30% review capacity, multimodal fusion prioritised 20 of 24 clinically significant segmentation failures (83.3%; 95% CI, 64.1–93.3%), compared with 17 of 24 (70.8%) using MRI alone. Fusion also achieved a higher area under the receiver operating characteristic curve than MRI alone (0.85 versus 0.78; difference, 0.07; 95% CI, 0.01–0.14). In a randomised crossover study of 4 clinicians, system assistance increased failure detection from 63.5% to 82.3% and correct review decisions from 71.9% to 83.8%, while adding 0.3 minutes per review. These findings support multimodal triage as a promising quality-control strategy for directing expert review of post-treatment glioma segmentations.

Keywords: trustworthy artificial intelligence, glioma, whole-tumour segmentation, clinical decision support, MRI–clinical–molecular fusion, prospective silent-mode validation, human-in-the-loop review

1 Introduction

Magnetic resonance imaging (MRI) is central to post-treatment assessment of diffuse glioma [1, 2]. Automated segmentation on multi-sequence MRI supports tumour-volume measurement, treatment planning and longitudinal assessment [3–5]. Curated benchmarks show strong average performance, but cohort-level overlap metrics do not identify which individual whole-tumour masks require expert correction. This challenge is greatest after treatment, when evolving tumour, treatment-related change and postoperative anatomy can overlap in appearance, while scanner, acquisition-protocol and population differences can shift an examination away from the model's development data [6, 7]. When expert review capacity is constrained, the clinically relevant task is therefore to prioritise the segmentations most likely to contain a consequential error.

Predictive uncertainty has emerged as a quality-control signal for medical artificial intelligence, including brain-tumour segmentation [8–12]. Prior brain-tumour work has developed voxelwise uncertainty maps, calibration analyses and case-level rankings on retrospective segmentation data. The next step is to determine whether uncertainty-based ranking can capture adjudicated clinically significant failures under limited review capacity and improve clinician decisions in prospective workflows. This requires operational endpoints: failures captured at a prespecified review capacity, failures remaining outside the priority set, and changes in clinician detection and review decisions. Studies of human–AI collaboration show that utility depends on reader decisions and workload, while real-world and silent-mode evaluations address performance in the intended workflow [13–16]. Comparable integrated evidence for post-treatment glioma segmentation is therefore needed.

In post-treatment glioma, MRI appearances are interpreted alongside treatment timing and molecular characteristics that define disease context [1, 2]. Together, these variables provide patient context that may complement imaging when estimating whether a segmentation warrants review. We therefore defined multimodal triage as combining an MRI-derived risk score with time since radiotherapy, isocitrate dehydrogenase (IDH) status and O^6-methylguanine-DNA methyltransferase (MGMT) promoter methylation status at the patient level, rather than treating the individual MRI sequences as separate clinical modalities. Multimodal, uncertainty-aware systems have shown promise in other clinical workflows [16, 17], motivating our test of whether patient context improves the ranking of post-treatment glioma segmentation failures beyond imaging alone.

To address this gap, we developed a quality-control system around a frozen BraTS segmentation backbone. The system combines complementary uncertainty signals into a voxelwise trust map, summarises the lowest-trust predicted tumour regions as a patient-level MRI risk score and fuses that score with the prespecified patient context to assign a review priority. It does not alter the automatic whole-tumour segmentation; it provides the trust map and priority score to guide expert review under a fixed review capacity.

We specified the fusion model in an independent 50-patient retrospective development cohort and locked its inputs, coefficients, calibration and operating rule before evaluation in a single-centre prospective silent-mode cohort of 80 consecutive patients.

Table 1 Public cohorts used for retrospective development and external technical validation. n_{full} denotes the source release and n_{used} the analysed units; repeated MU-Glioma-Post timepoints are clustered by patient.

Cohort	n_{full}	n_{used}	**Role**
BraTS-2023 GLI	1251	1251	in-distribution training + out-of-fold validation
BraTS-Africa (glioma)	95	95	primary full-cohort leakage-audited external test
UPenn-PDGM	611	509	unlabelled external-shift diagnostic
MU-Glioma-Post	596 timepoints (203 patients)	596	post-treatment external test and contextual comparator
Decathlon Task01	484	20	simulator parent pool only (n=20 parents)

Sources. BraTS-2023 GLI: Synapse SYN51514132. BraTS-Africa: TCIA (includes all BraTS-2023 SSA). UPenn-PDGM: TCIA (102/611 dropped at `dcm2niix`). MU-Glioma-Post: TCIA (*Scientific Data* 2025). Decathlon Task01: `medicaldecathlon.com`.

At a prespecified 30% review capacity, we compared multimodal fusion with MRI-only and clinical–molecular rankings by their ability to capture adjudicated clinically significant whole-tumour segmentation failures. A 4-clinician randomised crossover study separately tested whether displaying the final trust map and score improved failure detection and accept/edit/defer decisions, and evaluated confidence and review time. Public-cohort shift and simulated-mimic analyses provided supporting stress tests. Together, these analyses tested whether multimodal triage can direct limited expert review towards clinically significant segmentation errors and improve clinician review decisions.

2 Results

Results are presented in two stages. We first developed and technically validated the imaging trust map across four public glioma MRI cohorts, with a fifth public dataset used only to construct simulator-derived stress tests. We then evaluated the locked system prospectively in 80 consecutive adults and assessed its effect on review performance in a randomised crossover study with 4 clinicians.

2.1 Trust-map development and validation across public datasets

Public cohorts and model locking

The technical analysis comprised out-of-fold estimates for all 1,251 BraTS-2023 GLI cases and external estimates for all 95 leakage-audited BraTS-Africa cases and 596 MU-Glioma-Post timepoints from 203 patients. Supporting acquisition-shift and simulator analyses used 509 UPenn-PDGM cases and 20 Decathlon cases, respectively (Table 1). The cross-cohort audit found no strong matches and no patient-ID overlap between the external cohorts and either the frozen backbone-training or trust-map-development pools.

All five GLI validation folds selected the same interior mixing coefficient, $\alpha^{\star} = 0.6$. Mean validation ECE was 0.160 at the selected coefficient and 0.266 at the MC-Dropout corner ($\alpha = 1$), a gap of 0.106 ± 0.007. The mixture was then locked for the public external cohorts and the clinical study. Because the segmentation backbone remained frozen, whole-tumour Dice was unchanged across trust-map methods:

Table 2 Retrospective performance of the locked imaging trust map. GLI results are out of fold; BraTS-Africa and MU-Glioma-Post use the locked GLI-reference model. Expected calibration error (ECE) and Brier score are full-brain-mask correctness readouts; out-of-distribution area under the receiver operating characteristic curve (OoD-AUROC) compares perturbed with reference whole-tumour voxels. UPenn-PDGM is omitted because compatible harmonised labels were unavailable.

Method	Cohort	ECE ↓	Brier ↓	OoD-AUROC ↑
MC-Dropout [10]	BraTS-2023 GLI	0.492	0.250	0.746
	BraTS-Africa	0.479	0.250	0.664
	MU-Glioma-Post	0.473	0.245	0.650
Temperature-scaling [18]	BraTS-2023 GLI	0.440	0.226	0.708
	BraTS-Africa	0.479	0.250	0.580
	MU-Glioma-Post	0.318	0.171	0.697
SWAG [19]	BraTS-2023 GLI	0.450	0.224	0.738
	BraTS-Africa	0.469	0.245	0.603
	MU-Glioma-Post	0.479	0.248	0.508
MSP [20]	BraTS-2023 GLI	0.427	0.219	0.711
	BraTS-Africa	0.479	0.250	0.585
	MU-Glioma-Post	0.478	0.248	0.712
ODIN [21]	BraTS-2023 GLI	0.492	0.250	0.622
	BraTS-Africa	0.479	0.250	0.571
	MU-Glioma-Post	0.482	0.250	0.695
Energy [22]	BraTS-2023 GLI	0.492	0.250	0.624
	BraTS-Africa	0.479	0.250	0.677
	MU-Glioma-Post	0.482	0.250	0.642
ViM [23]	BraTS-2023 GLI	0.395	0.202	0.578
	BraTS-Africa	0.413	0.219	0.661
	MU-Glioma-Post	0.213	0.110	0.523
Reported trust-map mixture ($\alpha^{\star}$=0.6)	BraTS-2023 GLI	0.385	0.195	0.791
	BraTS-Africa	0.358	0.182	0.721
	MU-Glioma-Post	0.367	0.188	0.704

0.934 on GLI, 0.940 on BraTS-Africa and 0.857 on MU-Glioma-Post. The subsequent analyses therefore evaluated how well the trust map identifies and ranks unreliable masks.

Internal and external technical validation

Compared with MC-Dropout, the locked mixture reduced ECE from 0.492, 0.479 and 0.473 to 0.385, 0.358 and 0.367 on GLI, BraTS-Africa and MU-Glioma-Post, respectively (Table 2). Brier scores decreased from 0.250, 0.250 and 0.245 to 0.195, 0.182 and 0.188. OoD-AUROC increased from 0.746 to 0.791 on GLI, from 0.664 to 0.721 on BraTS-Africa and from 0.650 to 0.704 on MU-Glioma-Post, corresponding to a mean improvement of 0.052 across the three labelled cohorts.

The focused comparison in Figure 1 shows the fusion-rule selection result and the consistent OoD-detection advantage of the locked final mixture over MC-Dropout across the three labelled cohorts.

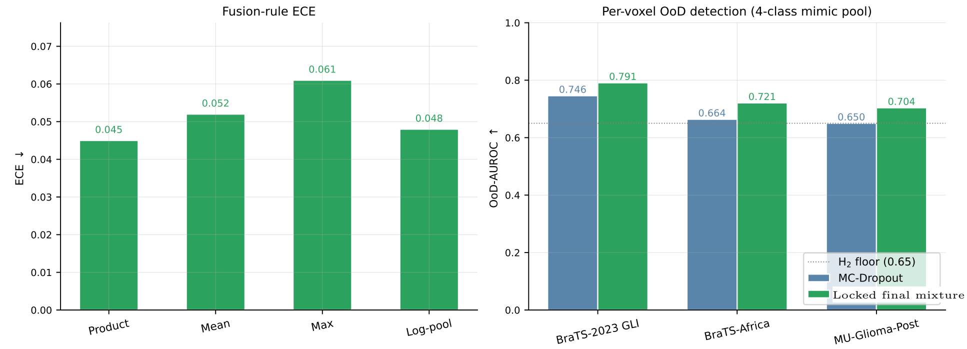

Fig. 1 Fusion-rule calibration and per-cohort OoD detection. (a) Expected calibration error (ECE) for product, mean, maximum and log-pooling applied to the three-mechanism core ($\alpha = 0$) on the GLI operator-selection pool. **(b)** Per-voxel out-of-distribution area under the receiver operating characteristic curve (OoD-AUROC) for MC-Dropout and the locked final mixture at $\alpha^\star = 0.6$ on GLI, BraTS-Africa and MU-Glioma-Post. UPenn-PDGM is omitted because compatible harmonised labels were unavailable.

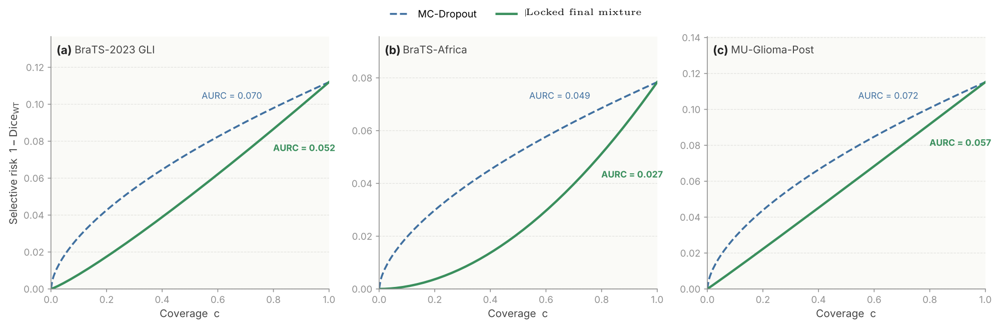

Fig. 2 Selective-prediction risk–coverage across public cohorts. The locked final mixture at $\alpha^\star = 0.6$ (solid green) is compared with MC-Dropout (dashed blue). Cases are retained from higher to lower confidence; risk is $1 - \mathrm{Dice}_{\mathrm{WT}}$, and lower curves are better.

The direction of improvement was consistent between out-of-fold GLI, the sub-Saharan African external cohort and the post-treatment external cohort. This consistency supported the prespecified use of the locked trust map as the MRI risk input for prospective review triage.

Selective prediction and cross-cohort transportability

When cases were retained from higher to lower trust, the final mixture produced lower risk–coverage curves than MC-Dropout on all three labelled cohorts (Figure 2). The paired AURC differences were -0.018 (95% CI, -0.032 to -0.005) on GLI, -0.022 (95% CI, -0.031 to -0.013) on BraTS-Africa and -0.015 (95% CI, -0.018 to -0.012) on MU-Glioma-Post. Thus, the locked mixture more effectively concentrated segmentation error among the cases selected for review.

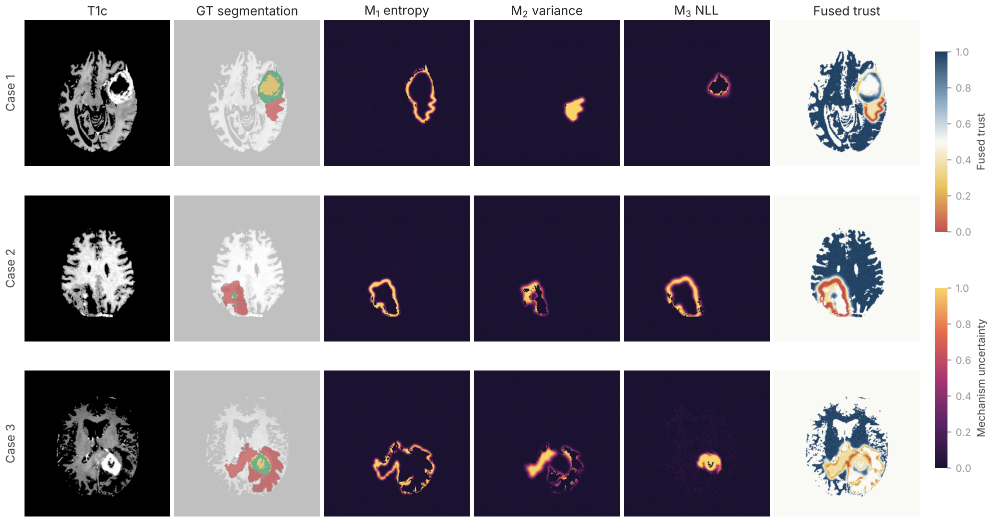

Fig. 3 Per-voxel uncertainty decomposition in three illustrative BraTS-2023 GLI cases. From left to right: T_{1c} slice, ground-truth segmentation, M_1 prototype-posterior entropy, M_2 ensemble predictive variance, M_3 flow negative log-likelihood and the fused trust map at $\alpha^\star = 0.6$. Warm colours denote greater uncertainty in the component panels; cool blue denotes higher trust in the fused map. This figure provides a qualitative spatial display; quantitative estimates use the full evaluation cohorts.

Spatial interpretation of the trust map

The risk–coverage results show that the final mixture ranks unreliable cases more effectively. The illustrative cases in Figure 3 show how the three component signals localise complementary spatial patterns and how their fusion produces a single trust overlay for clinical review.

Across the four glioma cohorts, channel-aligned representation similarity remained high (CKA, 0.89–0.95). Cohort-specific flows had higher held-out density fit on three of four cohorts, while off-diagonal calibration-transfer ratios ranged from 1.1× to 3.6× (median, approximately 2.1×). These diagnostics showed a shared representation structure while identifying calibration as the component most sensitive to local cohort conditions; none of the external estimates was retuned.

2.2 Prospective clinical validation and utility

Prospective silent-mode cohort and operational feasibility

The retrospective development cohort comprised 50 post-treatment glioma patients, including 15 with a clinically significant whole-tumour segmentation failure. After model locking, all 80 consecutively enrolled adults in the prospective silent-mode cohort completed evaluation.

Blinded expert adjudication identified 24 (30.0%) clinically significant whole-tumour segmentation failures and 56 segmentations without such a failure. The

Table 3 Prospective review-triage performance at the prespecified 30% review capacity. Fusion combines the locked MRI trust-map risk with time since radiotherapy, IDH status and MGMT promoter methylation. Each model selects its own 24 priority cases; failures outside priority review are counted within that model's corresponding 56 unselected cases.

Model	Failures captured	Failures outside priority review	AUROC (95% CI)	AUPRC (95% CI)
Clinical–molecular context	13/24 (54.2%)	11/56 (19.6%)	0.68 (0.55–0.80)	0.48 (0.34–0.63)
MRI risk score	17/24 (70.8%)	7/56 (12.5%)	0.78 (0.67–0.88)	0.64 (0.49–0.77)
MRI–clinical–molecular fusion	20/24 (83.3%; 95% CI, 64.1–93.3%)	4/56 (7.1%)	0.85 (0.76–0.93)	0.75 (0.60–0.87)

pipeline processed all 80 examinations without manual preprocessing, with a median end-to-end inference time of 2.6 minutes per examination.

Prospective multimodal review-triage performance

At the prespecified 24-of-80 (30%) review capacity, MRI–clinical–molecular fusion prioritised 20 of 24 clinically significant failures (83.3%; 95% CI, 64.1–93.3%), compared with 17/24 (70.8%) using MRI risk alone and 13/24 (54.2%) using clinical–molecular context alone. Fusion left four failures outside priority review, compared with seven for MRI risk and 11 for clinical–molecular context. The fusion–MRI capture difference was 12.5 percentage points (95% CI, −2.1 to 26.8).

Across all thresholds, fusion achieved an AUROC of 0.85 (95% CI, 0.76–0.93), compared with 0.78 (0.67–0.88) for MRI risk and 0.68 (0.55–0.80) for clinical–molecular context. The paired AUROC improvement over MRI risk was 0.07 (95% CI, 0.01–0.14). AUPRC followed the same ordering: 0.75 (95% CI, 0.60–0.87) for fusion, 0.64 (0.49–0.77) for MRI risk and 0.48 (0.34–0.63) for clinical–molecular context. The concordant AUROC and AUPRC gains showed that fusion improved failure ranking beyond the imaging trust map alone (Table 3).

Among 19 cases meeting the treatment-related-change reference, seven were supported by tissue and 12 by multidisciplinary consensus with longitudinal follow-up. Nine had a clinically significant whole-tumour segmentation failure; fusion prioritised eight at the same cohort-level quota, compared with six using MRI risk alone.

Clinician reader evaluation

All four clinicians completed 640 randomised crossover assessments, with 320 in each condition. Across the 96 reader–case opportunities per condition containing a clinically significant failure, detection increased from 61 (63.5%) without the combined display to 79 (82.3%) with it, an adjusted increase of 18.8 percentage points (95% CI, 8.7–28.8).

Correct accept, edit or defer decisions increased from 230/320 (71.9%) to 268/320 (83.8%), an adjusted increase of 11.9 percentage points (95% CI, 6.1–17.6). Among reader–case opportunities without a clinically significant failure, editing decreased from 36/224 (16.1%) to 22/224 (9.8%), an adjusted difference of −6.3 percentage points (95% CI, −11.7 to −0.8).

Median review time was 4.3 minutes without assistance and 4.6 minutes with assistance; the adjusted mean difference was 0.3 minutes (95% CI, 0.1–0.5). Mean confidence increased from 3.2 to 3.8 on a five-point scale, an adjusted difference of 0.6 (95% CI, 0.4–0.8). The combined display increased failure detection, correct review dispositions and confidence, with a 0.3-min increase in review time (Table 4).

Table 4 Randomised crossover evaluation with four clinicians. Control included multi-sequence MRI, patient context and the automatic mask; assistance additionally included the locked trust map and fusion score. Binary contrasts are adjusted percentage-point differences; time and confidence contrasts are adjusted mean differences.

Outcome	Control	Combined-display assisted	Adjusted difference (95% CI)
Clinically significant whole-tumour segmentation-failure detection	61/96 (63.5%)	79/96 (82.3%)	+18.8 pp (8.7–28.8)
Correct review disposition	230/320 (71.9%)	268/320 (83.8%)	+11.9 pp (6.1–17.6)
Editing among reference-no-significant-failure cases	36/224 (16.1%)	22/224 (9.8%)	−6.3 pp (−11.7 to −0.8)
Review time, descriptive median (min)	4.3	4.6	+0.3 adjusted mean (0.1–0.5)
Confidence, mean, 1–5	3.2	3.8	+0.6 adjusted mean (0.4–0.8)

3 Discussion

This study evaluated an uncertainty wrapper for a defined clinical decision: which post-treatment glioma whole-tumour segmentations should receive priority expert review when review capacity is constrained. In a prospectively accrued cohort analysed with a frozen model and operating rule, offline application of the prespecified 30% quota would have placed 20 of 24 clinically significant whole-tumour segmentation failures within the priority-review set (83.3%; 95% CI, 64.1–93.3%). 4 failures remained among the 56 cases outside the fusion priority set, compared with 7 for the MRI risk score alone and 11 for clinical–molecular context alone. Because the wrapper does not modify the underlying whole-tumour segmentation, it offers a practical way to reallocate fixed review capacity towards cases at greater risk of consequential error. The silent-mode design preserved existing clinical safeguards, and all cases remained subject to usual clinical review processes.

Fusion improved overall risk ranking over the MRI risk score alone, with an AUROC gain of 0.07 (95% CI, 0.01–0.14). At the prespecified 30% batch-relative quota, fusion captured three additional failures compared with MRI risk alone (20 versus 17); the corresponding 12.5-percentage-point difference was directionally favourable (95% CI, −2.1 to 26.8). Together, these findings show that patient-level fusion adds ranking information and can direct a fixed review capacity towards clinically important failures. The offline, cohort-level quota provides an operational starting point; future sites can adapt either a local batch policy or an absolute score threshold to their review capacity and tolerance for failures outside the priority set.

The multimodal contribution came from combining imaging with clinical and molecular variables rather than treating the four MRI sequences as separate modalities. T_1, contrast-enhanced T_1, T_2 and FLAIR constitute multi-sequence MRI within one imaging modality. The patient-level model additionally incorporated time from radiotherapy and the tissue-derived IDH and MGMT markers. The improvement over MRI risk alone indicates that imaging-derived uncertainty and clinical–molecular context contributed complementary information for review prioritisation. The fusion model remained parsimonious, comprising one MRI risk score, three non-imaging predictors and three prespecified missingness indicators.

The locked prospective design extends evaluation beyond purely retrospective uncertainty benchmarks. Model inputs, preprocessing and imputation rules, coefficients, calibration parameters and the review rule were specified before the first prospective index examination; all 80 consecutive eligible examinations were analysed; and the pipeline completed without manual preprocessing in a median of 2.6

minutes per examination. These observations support prospective silent-mode technical feasibility. Outputs were, however, withheld from treating clinicians, allowing the locked system to be evaluated without influencing care. Successful completion of this phase supports progression to workflow-integrated and interventional studies assessing clinical use, clinician adoption and patient-level impact.

Prior brain-tumour uncertainty studies have primarily evaluated calibration and retrospective case ranking on benchmark data [10–12]. The present study extends that work by evaluating a locked ranking rule against adjudicated clinically significant failures in a prospectively accrued cohort and by testing the final display in a randomised crossover reader study. This design also complements real-world and human–AI workflow evaluations in other imaging applications [13–16].

The randomised crossover reader evaluation provides complementary evidence that the display can influence clinician decisions. Combined single-map-and-fusion-score assistance increased clinically significant whole-tumour segmentation-failure detection by 18.8 percentage points and correct accept/edit/defer disposition by 11.9 percentage points. Editing frequency among reference-no-significant-failure cases was lower with assistance, consistent with more selective editing; this exploratory endpoint was not designed as a formal specificity measure. The adjusted mean review-time difference was 0.3 minutes per case. The study evaluated the clinically intended combined display of the final trust map and patient-level score; future dismantling studies can quantify their individual contributions and those of the component mechanisms. This pattern supports the display as a potential quality-control aid when modest additional review time is acceptable. The randomised, balanced and washout-controlled design across 4 clinicians provides internally controlled evidence of its effect, with evaluation in routine clinical service as the next step for confirming generalisability.

The mechanism-resolved channels and retrospective stress tests strengthen interpretation of the triage signal. Prototype geometry, ensemble disagreement and low-density feature support form the three-mechanism core; the single final map additionally blends the MC-Dropout reference channel. Public-cohort analyses assess technical transportability, while simulator-derived perturbations test whether controlled changes elicit distinguishable uncertainty patterns. Together, these analyses provide complementary evidence of technical transportability and mechanistic responsiveness, while diagnostic classification of recurrence or treatment-related change remains outside the intended use.

The results also define clear priorities for further translation. The parsimonious fusion model was developed in 50 patients with 15 failures; larger development samples can further stabilise its coefficients and calibration. The single-centre prospective cohort of 80 patients, including 24 clinically significant failures, provides initial locked-model clinical evidence; external multicentre cohorts can improve precision and evaluate site-specific calibration. In the secondary treatment-related-change subgroup, the pragmatic reference combined histopathology with longitudinal multidisciplinary consensus when tissue was unavailable, and future tissue-enriched cohorts can strengthen this analysis. The controlled reader study establishes an effect under standardised conditions, while workflow-integrated studies can evaluate use within clinical worklists and subsequent clinical actions. External prospective evaluation, subgroup and fairness analyses, local calibration and interventional implementation are the next steps towards deployment.

In conclusion, a locked MRI–clinical–molecular triage model improved overall ranking of glioma whole-tumour segmentations requiring expert review compared with the MRI risk score alone in a single-centre prospective silent-mode cohort, and the combined display improved review decisions relative to control in a controlled multi-reader study. These findings position the system as a review-prioritisation aid and support progression to external and interventional evaluation. Autonomous acceptance of segmentations and diagnostic classification remain outside its intended scope.

4 Methods

4.1 Imaging trust-map architecture

The imaging component attached three complementary uncertainty heads to features from frozen public glioma-segmentation backbones [24, 25] (Figure 4). Prototype-posterior entropy (M_1), ensemble predictive variance (M_2) and bottleneck-density departure under a RealNVP flow [26] (M_3) were converted to voxelwise confidence factors and multiplied to form a three-mechanism core. The final locked trust map mixed this core with an MC-Dropout confidence channel: $\widetilde{c}_v = 0.4(c_{1v}c_{2v}c_{3v}) + 0.6c_{4v}^{\mathrm{MCD}}$. The segmentation backbones remained frozen, so the trust layer ranked and localised uncertainty without altering the predicted mask.

For clinical triage, the voxelwise map was reduced to an examination-level MRI risk score using the lowest-trust decile within the predicted whole-tumour mask. The following subsections describe its fusion with time since radiotherapy, IDH status and MGMT promoter methylation, the prospective silent-mode evaluation and the reader crossover study. Detailed head definitions, objectives, hyperparameters and retrospective stress tests are reported in Supplementary Information.

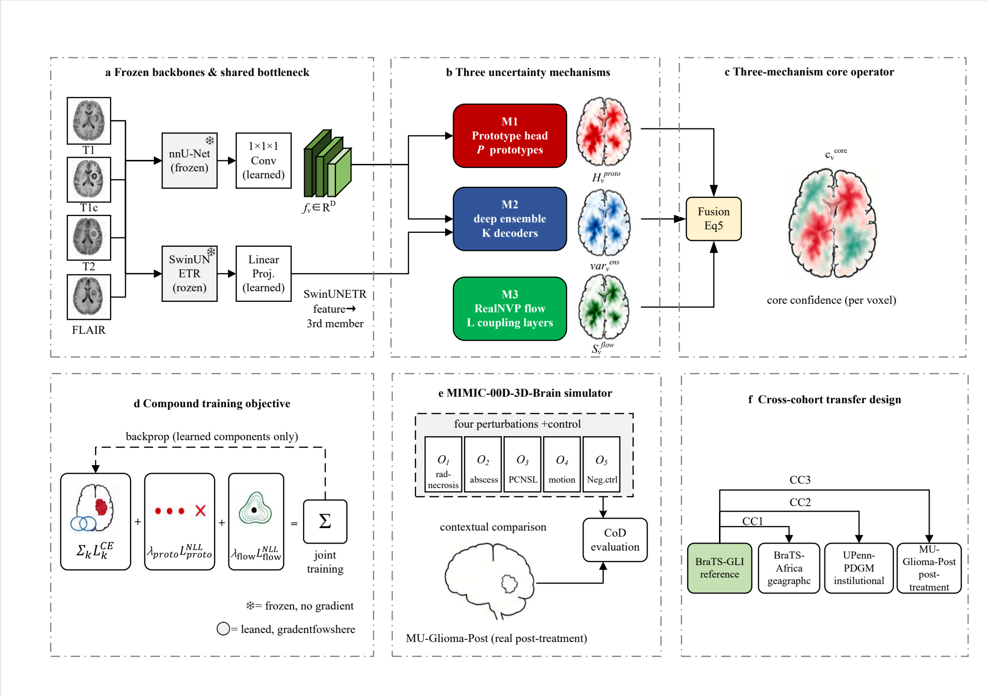

Fig. 4 Imaging trust-map architecture and technical-validation framework. (a) Four co-registered MRI sequences enter frozen nnU-Net and SwinUNETR backbones and a shared projected bottleneck. **(b)** Prototype geometry (M_1), ensemble disagreement (M_2) and density departure (M_3) generate complementary voxelwise uncertainty signals. **(c)** Their confidence factors form a product core that is mixed with MC-Dropout at the locked $\alpha^\star = 0.6$. **(d)** Training updates only the projections and uncertainty heads. **(e)** Simulator-derived perturbations support mechanism stress tests. **(f)** GLI cross-validation is used for selection and locking before unchanged external and prospective evaluation. The figure depicts the imaging component; patient-level MRI–clinical–molecular fusion is described in the accompanying Methods.

4.2 System overview and development sequence

The system was constructed in four stages. First, a frozen BraTS segmentation backbone generated the automatic whole-tumour mask. Second, prototype entropy, refinement-decoder disagreement and bottleneck-feature density were combined with MC-Dropout to produce a voxelwise imaging trust map. Third, the lowest-trust 10% of predicted tumour voxels were aggregated into a patient-level MRI risk score. Finally, this score was fused with time since radiotherapy, IDH status and MGMT promoter methylation to rank cases for expert review. The imaging component was developed on BraTS-2023 GLI and technically validated across public cohorts; the patient-level fusion models were fitted in an independent 50-patient retrospective cohort; and the complete locked system was evaluated in a 80-patient prospective silent-mode cohort and a randomised crossover reader study. Detailed imaging architecture, training and retrospective stress tests are reported in the Supplementary Methods and Figure 4.

4.3 Imaging trust-map architecture and patient-level MRI risk

The imaging predictor was derived only from the four co-registered MRI sequences. A three-mechanism core combined prototype entropy (M_1), refinement-decoder disagreement (M_2) and low support under the training bottleneck-feature density (M_3). If $c_{1v}, c_{2v}, c_{3v} \in [0, 1]$ denote the corresponding confidence factors, the core confidence was $c_v^{\text{core}} = c_{1v}c_{2v}c_{3v}$. The single final reported map then blended this core with an MC-Dropout reference-confidence channel:

$$\widetilde{c}_v = 0.4c_v^{\text{core}} + 0.6c_{4v}^{\text{MCD}} \in [0, 1].$$

Technical cross-validation was used only for out-of-fold estimation and hyperparameter selection. After those choices were fixed, the mechanism heads, flow, feature normalisation and NLL references were refitted once on all 1,251 BraTS-2023 GLI cases. Imaging calibration parameters were estimated only on the prespecified inner GLI calibration partition and then locked. The resulting single full-GLI-reference wrapper generated the external-cohort and prospective maps; prospective $\widetilde{c}_v$ was therefore neither an average of fold-specific maps nor selected using prospective outcomes. Lower final trust indicates greater concern. The automatic whole-tumour mask was inherited from the frozen backbone and was not modified by the overlay. Further architectural and implementation details are provided in the Supplementary Methods and Figure 4.

Let W_i be the predicted whole-tumour mask for case i and $N_{v,i} = |W_i|$. When $N_{v,i} > 0$, we set $k_i = \max\{1, \lceil 0.10N_{v,i} \rceil\}$ and defined case confidence C_i as the mean of the k_i lowest values of the final fused trust $\widetilde{c}_v$ within W_i. A fixed voxel-index order resolved an exact boundary tie; because tied voxels have the same trust value, this does not change the mean. When $N_{v,i} = 0$, we set $C_i = 0$. The clinical MRI risk score was $R_i = 1 - C_i$, so an empty prediction receives maximal risk $R_i = 1$ and higher values indicate greater review priority. The supporting selective-prediction analyses use the same aggregation in the confidence direction, retaining cases from higher to lower C_i.

4.4 Clinical–molecular fusion and model locking

The three prespecified non-imaging predictors were continuous elapsed time in months from completion of radiotherapy to the index MRI, IDH status (mutant coded 1; wild type coded 0) and MGMT promoter status (methylated coded 1; unmethylated coded 0). Unknown or indeterminate status was treated as missing. Non-missing continuous values were standardised using the development-cohort mean and standard deviation; a missing timing value was set to zero after standardisation and accompanied by a timing-missingness indicator. A missing binary value was set to zero and accompanied by its own missingness indicator. The indicator therefore distinguished missing status from wild type or unmethylated status.

We compared three patient-level ridge-logistic models: clinical–molecular context alone, the MRI risk score alone and MRI–clinical–molecular late fusion. The late-fusion design comprised one MRI risk score, the three non-imaging predictors and their three predictor-specific missingness indicators. Predictor values were restricted to information available at or before the index examination. Regularisation strength, coefficients and calibration parameters were determined only in the development cohort and frozen before prospective evaluation. Post-index histopathology, follow-up imaging, clinical outcomes and reader decisions were never model inputs.

4.5 Clinical study design and cohorts

We conducted a single-centre clinical-feasibility study at a tertiary referral hospital. The prespecified intended use was workload-constrained quality control: to rank post-treatment glioma whole-tumour segmentations for expert review. Autonomous mask acceptance and diagnostic classification of tumour recurrence or treatment-related change were outside scope. Throughout the clinical study, whole-tumour denotes the binary union of all BraTS tumour subregions produced by the frozen backbone; no clinical endpoint is claimed for an individual subregion. The study comprised an independent retrospective development cohort, a prospective silent-mode validation cohort and a randomised crossover reader evaluation.

Independent denotes patient- and examination-disjoint development and prospective cohorts, with no prospective-cohort data used for model fitting, calibration or operating-rule selection. *Consecutive* denotes screening all eligible patients in temporal order during the defined accrual window without selection on score, mask quality or reference outcome. *Prospective* denotes that the protocol, locked model and index-examination data collection were applied forward from the first prospective examination; the later six-month follow-up used only for the secondary aetiologic reference was not a model input.

The development cohort included 50 independent post-treatment glioma patients. The prospective cohort comprised 80 consecutive eligible adults, with one prespecified index examination per patient. Eligibility required an established diffuse glioma diagnosis, prior tumour-directed treatment and an examination containing T_1, T_{1c}, T_2 and FLAIR. The complete triage specification—predictor definitions, imputation rules, regularisation, coefficients, calibration and the review operating rule—was locked before the first prospective index examination. Silent-mode outputs were withheld from treating clinicians and did not alter care; the prospective evaluation therefore assessed ranking and operational feasibility rather than implementation in routine practice.

4.6 Segmentation-failure reference standard and treatment-related-change adjudication

Two distinct reference standards were used. The primary reference standard was clinically significant whole-tumour segmentation failure. Two board-certified neuroradiologists, blinded to the final MRI trust map, fusion score and review rank, independently reviewed each examination and, where needed, corrected the automatic whole-tumour mask. A failure was prospectively defined as an omission or commission for which correction changed the whole-tumour volume estimate relative to the automatic mask by at least 1 ml and at least 10%, or that both adjudicators judged capable of altering longitudinal tumour measurement or target-contour review. Disagreements were resolved by a third senior neuroradiologist.

The panel also assigned one reference disposition. *Accept* meant that no correction was judged necessary. *Edit* meant that correction was judged appropriate, irrespective of whether the error met the clinically significant failure threshold. *Defer* meant that the index examination alone was insufficient for a definitive disposition and additional information or specialist review was required. A reader disposition was correct only when it exactly matched this adjudicated category. A reference-no-significant-failure case could therefore still have an appropriate minor edit.

A separate aetiologic reference classified predominant treatment-related change. Histopathology was used when tissue was available; otherwise multidisciplinary consensus incorporating at least six months of imaging and clinical follow-up was used. This label did not define the primary segmentation-failure endpoint and supported only the prespecified descriptive subgroup analysis. Baseline IDH and MGMT results were model predictors, whereas post-index tissue and follow-up used for aetiologic adjudication were not used for model fitting, calibration, operating-rule selection or updating.

4.7 Controlled randomised crossover reader evaluation

Four clinicians who did not participate in reference adjudication completed a randomised two-period crossover evaluation. In the control condition, readers viewed the four MRI sequences, the clinical–molecular information available at the index examination and the automatic whole-tumour segmentation. In the assisted condition, they viewed the same information plus the colour-coded single final fused voxelwise MRI trust map and the patient-level MRI–clinical–molecular fusion score. Component M_1, M_2, M_3 and MC-Dropout maps were neither displayed nor separately tested. The map and score were presented as local and case-level review cues, respectively, and not as diagnoses of recurrence or treatment-related change. Each reader assessed all 80 cases in both conditions; AB/BA order was balanced, case order was rerandomised and the washout period was at least four weeks. Readers recorded clinically significant whole-tumour segmentation-failure presence, accept/edit/defer disposition, confidence on a five-point scale (1 = very low; 5 = very high) and review time from opening the complete case to submitting the response.

4.8 Endpoints and statistical analysis

The prespecified offline primary endpoint was the proportion of clinically significant whole-tumour segmentation failures falling within a batch-relative priority-review quota. For each locked model, the 80 completed prospective cases were ranked by the unrounded full-precision score, and exactly the highest-risk 24 cases (30%) were designated for priority review. An exact tie crossing the quota boundary, if present, was resolved by a deterministic deidentified case-identifier order frozen before outcome adjudication. Each model therefore left 56 cases outside its own priority set; these three sets can differ in membership. This count is numerically equal to the 56 reference-no-significant-failure cases, but the sets are not interchangeable because an outside-priority set can contain failures. The quota was not an absolute score threshold for sequential or streaming deployment. Cases were not actually routed during silent-mode evaluation.

The primary fusion capture proportion used a two-sided 95% Wilson score interval without continuity correction. AUROC, AUPRC and between-model patient-level contrasts used 2,000 paired nonparametric patient bootstrap resamples. For quota-specific capture contrasts, each resample was reranked and its highest-risk 30% reselected; duplicate resampled cases retained distinct stable occurrence indices for deterministic boundary handling.

Within the reader study, clinically significant failure detection, exact accept/edit/defer disposition, and editing among reference-no-significant-failure cases, confidence and review time were ordered for interpretation as primary, secondary and exploratory outcomes, respectively. Failure-detection models used the 192 reference-positive reader–case assessments (96 per condition); editing-frequency models used the 448 reference-no-significant-failure assessments (224 per condition); disposition, time and confidence used all 640 assessments (320 per condition). Binary outcomes were analysed with mixed-effects logistic regression containing assistance condition, study period and sequence as fixed effects and crossed random intercepts for reader and case. Adjusted absolute percentage-point differences were obtained by average marginal standardisation. Review time and confidence used linear mixed models with the same fixed effects and clustering structure; reported contrasts are adjusted mean differences, and a positive time contrast denotes longer review with assistance. All reader-model 95% confidence intervals used 2,000 parametric bootstrap draws. These controlled outcomes do not measure care delivery or patient outcomes. The study was not powered to demonstrate changes in treatment, progression-free survival or overall survival.

4.9 Ethics approval and consent to participate

The study protocol, including the retrospective development cohort, prospective silent-mode validation and reader study, was approved by the Institutional Review Board of the Second Hospital of Hebei Medical University (approval No. 2025-R725) and was conducted in accordance with the Declaration of Helsinki. The requirement for informed consent was waived for the retrospective cohort because only de-identified routinely acquired data were analysed. Written informed consent was obtained from participants enrolled in the prospective cohort. Algorithm outputs were not returned to treating clinicians and did not affect clinical management. All cases used in the reader study were de-identified before review.

4.10 Consent for publication

Consent for publication was not required because no identifiable individual participant data are reported.

Declarations

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Data availability

De-identified patient-level data are not publicly available because they contain sensitive clinical and molecular information and are governed by institutional ethics approval. Aggregate data and derived analysis outputs are available from the corresponding author upon reasonable request, subject to institutional review and a data-use agreement. Public retrospective cohorts used in this study are available from their original repositories.

Code availability

Code for preprocessing, trust-map inference, MRI–clinical–molecular fusion and statistical analysis is available from the corresponding author upon reasonable request.

Protocol and analysis availability

The prespecified study protocol and statistical analysis plan are available from the corresponding author upon reasonable request.

Author contributions

D.Z., J.L., C.T. and C.M. conceived and designed the study. D.Z., J.L., X.X., M.L., G.M., Y.L. and Y.Q. curated the data. C.T. and C.M. implemented the methods and performed the analyses. C.Z., D.W. and J.W. contributed clinical interpretation. X.L., Q.L. and Z.C. supervised the work. D.Z. and J.L. drafted the manuscript. All authors critically revised and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Supplementary Information

The following material reports detailed imaging methods, public-cohort procedures and retrospective technical stress tests that support, but do not define, the clinical review-triage claim.

S1 Supporting imaging architecture and retrospective evaluation

The clinical design, trust-map-derived MRI risk score and MRI–clinical–molecular fusion were specified above. The supporting imaging component comprised a three-mechanism core attached to a frozen public 3D glioma segmentation backbone, together with an MC-Dropout reference-confidence channel used in the reported mixture. The three core mechanisms (M_1, M_2, M_3) are operational proxies rather than identifiable causal sources: prototype geometry (M_1), ensemble disagreement (M_2) and low-density departure under the training bottleneck-feature distribution (M_3). This framing is informed by the Hüllermeier–Waegeman taxonomy [8], but we do not claim that any channel uniquely identifies an underlying uncertainty type.

We use *three-mechanism core* for $c_1c_2c_3$ and *final reported trust map* for $\widetilde{c}_v = 0.4(c_{1v}c_{2v}c_{3v}) + 0.6c_{4v}^{\mathrm{MCD}}$. Unless an analysis is explicitly labelled as an $\alpha = 0$ core ablation, the imaging trust map—including the map reduced to the clinical MRI risk score—refers to this four-channel reported mixture at $\alpha^\star = 0.6$. Its clinical scope is review prioritisation for predicted whole-tumour masks, not confidence in an individual tumour subregion or an aetiologic diagnosis.

Figure 4 depicts the imaging-only architecture, compound training objective, simulator-derived stress-test set and cross-cohort technical analyses. These retrospective analyses characterise calibration, selective-prediction behaviour, mechanism complementarity and transportability; they do not estimate prospective clinical effectiveness. The following subsections build the core and final per-voxel MRI trust quantities from the shared bottleneck upward and then specify the supporting simulator, transportability diagnostics and training objective.

S2 Frozen backbones and shared bottleneck

Our primary backbone was the top-of-leaderboard public BraTS-2021 nnU-Net checkpoint [24], retained verbatim with all parameters frozen. We chose this checkpoint because its high baseline Dice ensures that any operating-point degradation observed in our results is not attributable to a weak segmentation backbone. We added a secondary frozen backbone, SwinUNETR [25], as a source of ensemble diversity from a different inductive bias. SwinUNETR contributed only through its deepest-stage feature, which entered the pipeline through the third member of the deep ensemble; the reference segmentation prediction was driven by the primary nnU-Net throughout, so by construction backbone Dice is held constant across all uncertainty configurations.

Both backbones consumed the four co-registered MRI sequences $\{T_1, T_{1c}, T_2, \mathrm{FLAIR}\}$ at fixed isotropic resolution. Let $\boldsymbol{f}_v \in \mathbb{R}^D$ denote the bottleneck feature at voxel v, where D is the dimension of a learned $1 \times 1 \times 1$ convolution applied to the deepest-stage feature map. This projection is the only learned component shared between backbone and uncertainty heads; everything downstream of $\boldsymbol{f}_v$ — the three uncertainty mechanisms, the fusion operator, the identifiability sketch — operates on this single feature space, enabling the common feature-space comparison in the Channel-recovery sketch section. During cross-validation, per-channel means $\boldsymbol{m}^{(r)}$ and standard deviations $\boldsymbol{s}^{(r)}$ of $\boldsymbol{f}_v$ were estimated only from in-distribution foreground voxels in GLI training fold r and frozen for its validation fold. After hyper-parameters were selected, final statistics $\boldsymbol{m}_{\mathrm{GLI}}, \boldsymbol{s}_{\mathrm{GLI}}$ were estimated from all 1,251 GLI cases and locked with the single full-data wrapper. In either setting the common normalised coordinate was $\boldsymbol{z}_v = (\boldsymbol{f}_v - \boldsymbol{m})/\boldsymbol{s}$, componentwise; a zero-variance channel used unit scale. External and prospective cases used only the final full-GLI statistics.

S3 Three-mechanism uncertainty stack

Each mechanism operates on $\boldsymbol{f}_v$ to produce a per-voxel uncertainty signal (Figure 4(b)). We define the three signals in turn—M_1 prototype, M_2 ensemble, and M_3 flow—then specify the per-voxel fusion operator that combines them.

Prototype-entropy head (M_1).

We harvested a bank of P prototypes $\{\boldsymbol{\mu}_p \in \mathbb{R}^D\}_{p=1}^P$ by k-means on $\boldsymbol{z}_v$ embeddings of the in-distribution training partition, with P chosen as an upper bound on the morphological diversity expected within glioma sub-structures. The prototype centres $\boldsymbol{\mu}_p$ remained frozen after k-means initialisation. To provide a goodness-of-fit log-likelihood for the prototype-GMM feature NLL training term in Eq. (S8), we attached a per-prototype Gaussian-mixture density with diagonal covariance [27]; of this density only the per-prototype diagonal covariances and the per-prototype mixing weights are trained, with the centres held at the k-means solution.

For *inference-time uncertainty signalling*, we did not use this mixture density directly: the trained GMM density enters the loss of Eq. (S8) only, while M_1 at inference uses the cosine-softmax posterior below. Cosine separation is scale-invariant, but prototypes and queries must still share coordinates; we therefore used the same $\boldsymbol{z}_v$ space for both and defined the prototype-class posterior at voxel v as

$$p(p \mid \boldsymbol{z}_v) \;=\; \mathrm{softmax}_p(\tau \cdot \cos(\boldsymbol{z}_v, \boldsymbol{\mu}_p))\,, \tag{S1}$$

where τ is a cosine temperature controlling the entropy regime: small τ collapses the posterior toward uniform, large τ saturates onto the nearest prototype. The M_1 signal at voxel v is the single scalar posterior entropy across the P assignments in Eq. (S1),

$$H_v^{\mathrm{proto}} \;=\; -\sum_{p=1}^{P} p(p \mid \boldsymbol{z}_v)\, \log p(p \mid \boldsymbol{z}_v). \tag{S2}$$

It lies in $[0, \log P]$ before per-volume normalisation and is not the GMM likelihood used only in the training loss.

Deep ensemble (M_2).

We trained K refinement decoder heads on top of the frozen encoder. The choice of K followed the deep-ensemble guidance of Lakshminarayanan et al. [28] that a small K already captures most of the gain over a single network; member diversity came from independent random seeds. The third member additionally consumed the SwinUNETR deepest-stage feature through a learned linear projection. Let $\hat{y}_v^{(k)} \in [0,1]$ denote member k's scalar whole-tumour foreground probability at voxel v, rather than a hard label or a vector of class probabilities. The M_2 signal is the scalar per-voxel variance of those K probabilities,

$$\mathrm{Var}_v^{\mathrm{ens}} \;=\; \frac{1}{K}\sum_{k=1}^{K}\big(\hat{y}_v^{(k)} - \bar{y}_v\big)^2, \qquad \bar{y}_v = \tfrac{1}{K}\sum_{k=1}^{K}\hat{y}_v^{(k)}, \tag{S3}$$

which is a Monte Carlo proxy for the posterior predictive variance under the deep-ensemble interpretation [28, 29].

Z-scored RealNVP density head (M_3).

For density-based out-of-distribution detection we used a normalising flow [26, 30] consisting of L affine-coupling layers with alternating channel-wise binary masks; each coupling block was parameterised by a multi-layer perceptron of hidden width d_h (we use d_h rather than H to avoid collision with the prototype entropy H^{proto}). We used $L = 8$ as specified in the Implementation and statistical plan section. Flow inputs were the normalised features $\boldsymbol{z}_v$ defined above: fold-specific statistics were used for out-of-fold GLI evaluation, and the locked full-GLI statistics were used otherwise. This normalisation mitigates the documented covariate-shift failure mode of RealNVP on unnormalised features reported by Kirichenko et al. [31]. The flow is an unconditional density model over this normalised bottleneck feature space; no class or cohort label is supplied as an additional condition.

Let $\mathrm{NLL}(\boldsymbol{z}_v)$ denote negative log-likelihood under the flow density, in natural-log nats per D-dimensional voxel feature, and let $\mu_{\mathrm{ID}} = \mathbb{E}_{\mathrm{ID}}[\mathrm{NLL}]$ be its mean over the applicable GLI reference feature pool. The standalone detector uses

$$S_v^{\mathrm{flow}} \;=\; \mathrm{NLL}(\boldsymbol{z}_v) + \big|\mathrm{NLL}(\boldsymbol{z}_v) - \mu_{\mathrm{ID}}\big|, \tag{S4}$$

which equals $2\,\mathrm{NLL}(\boldsymbol{z}_v) - \mu_{\mathrm{ID}}$ when $\mathrm{NLL}(\boldsymbol{z}_v) \geq \mu_{\mathrm{ID}}$ and μ_{ID} otherwise. It therefore amplifies low-density departures but is constant below the ID mean; it does not progressively detect atypically high likelihood. Raw NLL is the second scalar and supplies the one-sided density factor in the core fusion below.

Three-mechanism core operator.

We aggregated the three mechanism signals into a per-voxel core confidence by

$$c_v^{\mathrm{core}} = (1 - \tilde{H}_v^{\mathrm{proto}}) \cdot (1 - \widetilde{\mathrm{Var}}_v^{\mathrm{ens}}) \cdot \sigma(-\rho\,(\mathrm{NLL}_v - q_\beta))\,, \tag{S5}$$

where $\tilde{H}_v^{\text{proto}}$ and $\widetilde{\text{Var}}_v^{\text{ens}}$ are min–max normalised to $[0,1]$ over the non-padding brain-mask voxels of each volume. If a tensor has zero range, its normalised uncertainty is set to zero. The low-percentile reference q_β and μ_{ID} are estimated from the same applicable GLI reference feature pool. The slope $\rho > 0$ is fitted on an inner training calibration subset by minimising voxel-correctness Bernoulli NLL and then frozen. The three core confidence factors are $c_1 = 1 - \tilde{H}_v^{\text{proto}}$, $c_2 = 1 - \widetilde{\text{Var}}_v^{\text{ens}}$ and $c_3 = \sigma[-\rho(\text{NLL}_v - q_\beta)]$. Increasing NLL lowers c_3, whereas decreasing NLL raises it; a high c_3 cannot override a low c_1 or c_2 because the factors multiply. Thus Eq. (S5) defines the monotone-AND core confidence $c_v^{\text{core}} = c_1 c_2 c_3$, not the final reported trust map.

For the operator-selection ablation, product, arithmetic mean and maximum are respectively $\prod_{j=1}^{3} c_j$, $\frac{1}{3}\sum_{j=1}^{3} c_j$ and $\max_j c_j$. Equal-weight log-pooling is the geometric mean $\exp\{\frac{1}{3}\sum_{j=1}^{3}\log[\max(c_j, 10^{-6})]\}$. These operators are compared only for the $\alpha = 0$ three-mechanism core.

S4 MC-Dropout corner containment

We first motivate why MC-Dropout is the corner the family must recover, rather than an arbitrary baseline. MC-Dropout is the de facto single-channel uncertainty estimator for medical segmentation and the standard comparator in the segmentation-uncertainty literature [32], so making the fusion family contain it as a corner reframes the trust map as a backward-compatible extension of the MC-Dropout baseline rather than positioning it as a parallel replacement. This corner-containment has a concrete by-construction consequence: because $\alpha{=}1$ lies in the search grid, the val-fold maximiser $\alpha^\star$ satisfies $J(\alpha^\star) \geq J(\alpha{=}1)$ on the validation fold by construction, where J is the paired calibration–ranking objective formalised below. External-cohort transfer was evaluated empirically and was not inferred from this inequality.

With this motivation, we extend the fusion of Eq. (S5) to a one-parameter family that contains MC-Dropout as a corner. During stochastic inference, dropout with probability 0.2 was enabled after each decoder refinement block; all weights and normalisation layers remained frozen. If $p_v^{(t)}$ is the whole-tumour foreground probability on pass t, we defined

$$\bar{p}_v = \frac{1}{T}\sum_{t=1}^{T} p_v^{(t)}, \qquad V_v^{\text{MCD}} = \frac{1}{T}\sum_{t=1}^{T}\left(p_v^{(t)} - \bar{p}_v\right)^2, \qquad c_{4v}^{\text{MCD}} = 1 - \widetilde{V}_v^{\text{MCD}}. \tag{S6}$$

Here $\widetilde{V}_v^{\text{MCD}}$ is min–max normalised within the non-padding brain mask of each volume; a zero-range variance map is assigned normalised uncertainty zero. The value $T = 8$ is specified in the Implementation and statistical plan section. Let c_1, c_2, c_3 denote the three core mechanism confidences. We parameterise the trust map by a mixing coefficient $\alpha \in [0,1]$:

$$\widetilde{c} = (1-\alpha)\,(c_1\,c_2\,c_3) + \alpha\,c_4^{\text{MCD}}, \quad \alpha = 1 \Longrightarrow \widetilde{c} = c_4^{\text{MCD}}. \tag{S7}$$

The three-mechanism core is recovered at $\alpha = 0$, and the family reduces to MC-Dropout at $\alpha = 1$. On each GLI validation fold we maximise $J(\alpha) = -\text{ECE}_{\text{val}}(\widetilde{c}|\alpha) +$

AUROC$_{\mathrm{val}}(1-\widetilde{c}|\alpha)$, where OoD is the positive class. ECE$_{\mathrm{val}}$ is computed separately for each validation patient over non-padding brain-mask voxels using the voxel-correctness target and 15 fixed bins, then averaged with equal patient weight. The Brier sensitivity analysis uses the same patient-equal support. For the AUROC term, reference whole-tumour voxels from O_1–O_4 are positives and those from matched O_5 pass-through renders are controls; padding and non-tumour background are excluded. Simulator parents are kept wholly within a training or validation fold. Because the finite grid includes $\alpha = 1$, its validation maximiser satisfies $J(\alpha^\star) \geq J(1)$ on that fold; this is not a test-cohort guarantee. All folds select $\alpha^\star = 0.6$. With all choices fixed, the mechanism heads, flow, normalisation and NLL references are refitted on the complete 1,251-case GLI pool; scalar calibration parameters are estimated only on the prespecified inner GLI calibration partition. The resulting single locked model is used outside GLI and yields the final reported map $\widetilde{c}_v = 0.4c_v^{\mathrm{core}} + 0.6c_{4v}^{\mathrm{MCD}}$.

For both technical and clinical case reduction, let W_i be the predicted whole-tumour mask and let $k_i = \max\{1, \lceil 0.10|W_i|\rceil\}$ when $|W_i| > 0$. Case confidence C_i is the mean of the k_i lowest final-map values in W_i; $C_i = 0$ for an empty mask. Exact voxel-value ties use fixed voxel-index order. Technical selective prediction retains cases from high to low C_i; the clinical MRI risk is exactly $R_i = 1 - C_i$. The MC-Dropout comparator uses the same spatial aggregation on its own confidence map.

The following information-set bound motivates evaluating the combined signals but does not guarantee improvement for the fixed one-parameter mixture.

Proposition S1 (Information-set monotonicity)**.** *Let $Z \in \{0,1\}$ be the OoD indicator, $\mathcal{F}_{\mathrm{MCD}} = \sigma(c_4^{\mathrm{MCD}})$ and $\mathcal{F}_{\mathrm{all}} = \sigma(c_1, c_2, c_3, c_4^{\mathrm{MCD}})$. For any proper binary scoring rule L, define the unrestricted Bayes risk $R^\star(\mathcal{F}) = \inf_{g:\mathcal{F}\to[0,1]} \mathbb{E}[L(g, Z)]$. Then*

$$R^\star(\mathcal{F}_{\mathrm{all}}) \leq R^\star(\mathcal{F}_{\mathrm{MCD}}).$$

Proof sketch. $\mathcal{F}_{\mathrm{MCD}} \subseteq \mathcal{F}_{\mathrm{all}}$, so every $\mathcal{F}_{\mathrm{MCD}}$-measurable predictor is also $\mathcal{F}_{\mathrm{all}}$-measurable; taking the infimum over the larger function class cannot increase risk. □

This bound concerns unrestricted Bayes-optimal rules on two information sets. It neither asserts strict improvement nor establishes the performance of the scalar α-mixture. That family is instead supported by the finite-grid validation comparison and external estimates reported below.

S5 Channel-recovery sketch

We accompany the three-mechanism core with a *channel-recovery sketch* that states the operating conditions under which the signals are expected to respond to different perturbations. The observables are prototype-class boundary geometry, ensemble-member disagreement and one-sided low-density departure in the shared bottleneck. The argument rests on three explicit assumptions: (i) a non-collapsed, separated prototype bank and non-degenerate cosine-softmax posterior for M$_1$ [27]; (ii) the deep-ensemble interpretation of M$_2$ as a Monte Carlo posterior-predictive proxy [28]; and (iii) a stable reference flow-density model for M$_3$ conditional on the applicable

GLI-reference bottleneck z-score normalisation. Under these assumptions, the mechanisms are plausibly sensitive to distinct operational channels, with residual ambiguities including prototype-label permutation in M_1 and invertible reparameterisation of the flow base distribution in M_3 [33]. This is a design rationale, not an identifiability theorem. In the synthetic check, the planted directions are denoted a_{proto}, a_{ens} and a_{flow}; the last represents only a low-density shift. The sketch is evaluated empirically in the factorial-ablation and synthetic-recovery analyses.

The sketch is supported empirically by a small synthetic-data recovery check; the empirical witness on glioma cohorts is the factorial-ablation distinctness contrast reported in the Mechanism-resolved stress-test evidence section.

S6 MIMIC-OOD-3D-Brain simulator

We developed MIMIC-OOD-3D-Brain as a simulator-derived stress-test set (Figure 4(e)). It contains four planted perturbation conditions (O_1–O_4) and one unmodified pipeline pass-through control (O_5), generated with non-deep-learning operators on parent cases assigned wholly to one side of each cross-validation split. We do not treat O_5 as a fifth OoD source or as a strict model-ID cohort because 20 of the 100 parents come from Decathlon. The four perturbations cover radiation-necrosis-like morphology, abscess-like morphology, a PCNSL-like signature and motion artefact, but do not exhaust the broader look-alike spectrum.

O_1 simulates the post-treatment radiation-necrosis confusion mode with a thin T_{1c}-bright rim around a central necrotic core and FLAIR suppression in the peri-rim region, anchored to the descriptors of Schaff and Aboian [34]. O_2 simulates an abscess-like rim-enhancing lesion with a low-T_2 cavity, high-T_2 capsule and surrounding FLAIR-hyperintense oedema [35, 36]. O_3 simulates a uniformly enhancing, non-necrotic, FLAIR-homogeneous PCNSL-like signature [37]. O_4 applies a motion-artefact augmentation mapped qualitatively to the Krupa reference [38]. O_5 re-renders the unmodified parent through the same pipeline and therefore controls only for pipeline-induced effects.

Sampling was crossed at the parent level. Each of 100 simulator parents contributed one low-, medium- and high-severity render to each O_1–O_4 condition, giving 300 derived volumes per condition. O_5 likewise contributed three pass-through renders per parent to preserve matched replication; those three indices denote replicates, not severity. All simulator intervals and resampling use the parent as the clustering unit, so within-parent renders are not treated as independent cases. Fold-level holdout applied only during cross-validation. The final full-GLI wrapper had seen the 80 unmodified GLI parent scans during refitting but had never seen any derived perturbation volume; its simulator readout is consequently a perturbation-level, not parent-level, holdout. The 20 Decathlon parents were never part of the GLI refit. Headline OoD analyses use O_1–O_4 and exclude O_5 unless a pass-through-control comparison is explicitly stated.

The simulators are deterministic and seeded. Rim morphology is generated by anisotropic Perlin-modulated dilation with foreground-balanced soft-intensity blending, and intra-lesion intensity variability is controlled by a calibrated noise scale tuned to real glioma lesion descriptors. MU-Glioma-Post [6] provides a contextual post-treatment comparison, not a paired synthetic–real anchor or a patient-distribution surrogate.

S7 Cross-cohort transfer design

The headline external evaluation used a single locked reference-flow design. After cross-validation fixed all hyperparameters, the mechanism heads, flow, z-score statistics and NLL reference quantities were refitted once on all 1,251 BraTS-2023 GLI cases. Scalar calibration parameters were estimated only on the prespecified inner GLI calibration partition, locked, and then applied unchanged to BraTS-Africa, UPenn-PDGM [4] and MU-Glioma-Post [6]. No external-cohort flow refit enters the reported trust maps or clinical MRI score.

Separately, we conducted three supporting transportability diagnostics (Figure 4(f)). The universal-versus-per-cohort comparison contrasts a pooled flow with cohort-specific flows using patient-disjoint fit and evaluation partitions; longitudinal timepoints remain grouped by patient. This diagnostic asks whether site adaptation could improve descriptive density fit and does not replace the locked GLI-reference flow. The calibration-transfer matrix fits a temperature-only calibrator on a source-cohort fit partition among GLI, BraTS-Africa and MU-Glioma-Post and evaluates it on the disjoint target partition. Diagonal cells are likewise fit/evaluation separated. Transfer ECE pools all non-padding brain-mask voxels in the target evaluation partition and uses the same whole-tumour voxel-correctness target and 15 bins as the headline ECE. These source-specific calibrators do not retune the headline evaluation.

The third diagnostic is channel-aligned linear CKA. For cohort c, let $X_c \in \mathbb{R}^{D\times n_c}$ contain patient-level spatially pooled bottleneck features, with the same locked projected-bottleneck channel in each row. The learned $1\times 1\times 1$ projection is applied unchanged across cohorts. For each examination, spatial pooling is the arithmetic mean over non-padding brain-mask voxels; longitudinal examination vectors are then averaged within patient. We compare the centred $D \times D$ channel Gram matrices $K_c = X_c X_c^\top$ using $\mathrm{CKA}(a,b) = \langle HK_aH, HK_bH\rangle_F/(\|HK_aH\|_F\|HK_bH\|_F)$, where $H = I_D - D^{-1}\mathbf{1}\mathbf{1}^\top$. This channel-aligned orientation permits unequal, unpaired cohort sample counts because the shared observations are the $D = 256$ aligned channels, not patients. The PICTURE pathology representation [37] has a different channel basis and dimension and is therefore neither inserted into this matrix nor quantitatively compared with it.

S8 Compound training objective

Our training objective decomposes additively across the three mechanism heads (Figure 4(d)):

$$\mathcal{L} \;=\; \sum_{k=1}^{K} \mathcal{L}_k^{\mathrm{CE}}\big(\hat{y}^{(k)}, y\big) \;+\; \lambda_{\mathrm{proto}}\,\mathcal{L}_{\mathrm{proto}}^{\mathrm{NLL}} \;+\; \lambda_{\mathrm{flow}}\,\mathcal{L}_{\mathrm{flow}}^{\mathrm{NLL}}. \tag{S8}$$

Here K matches the deep-ensemble member count introduced above, so the sum is over the same K refinement heads; $\mathcal{L}_k^{\mathrm{CE}}$ is the per-voxel cross-entropy of ensemble member k on the downsampled segmentation target, $\mathcal{L}_{\mathrm{proto}}^{\mathrm{NLL}}$ is the prototype-GMM feature negative log-likelihood

$$\mathcal{L}_{\mathrm{proto}}^{\mathrm{NLL}} = -\sum_{v \in \mathcal{V}_{\mathrm{train}}} \log\left[\sum_{p=1}^{P} \pi_p \mathcal{N}\big(\boldsymbol{z}_v; \boldsymbol{\mu}_p, \mathrm{diag}(\boldsymbol{\sigma}_p^2)\big)\right],$$

where π_p and $\boldsymbol{\sigma}_p^2$ are the trained mixture weights and diagonal variances, and $\mathcal{L}_{\mathrm{flow}}^{\mathrm{NLL}}$ is the flow negative log-likelihood. The loss is written additively to identify the mechanism-specific terms; it should not be interpreted as complete parameter separability. Mechanism-specific parameters were optimised against frozen-backbone features, while the learned projection into $\boldsymbol{f}_v$ was shared by the heads that consumed that representation. Gradients were restricted to the learned projection and the relevant mechanism head and did not enter the frozen backbone. Consequently, backbone Dice remains mechanically constant across configurations. All hyperparameter values and training-schedule details (including $D, P, K, L, \tau, \lambda_{\mathrm{proto}}, \lambda_{\mathrm{flow}}$) are specified in the Implementation and statistical plan section.

S9 Public cohorts, splits, and leakage protocol

Table 1 lists the five public cohorts used only for retrospective imaging-component development or technical validation. The independent clinical development and prospective cohorts are described in the clinical Methods and are not included in this table.

- BraTS-2023 GLI [4, 39] is the in-distribution training pool. We use the full 1,251-case release to train and select the operating point for M_1–M_3 under 5-fold dedup-aware patient-level cross-validation (∼1,000 train / ∼250 val per fold). The held-out cohorts are evaluated on their full leakage-audited patient pools, with label-dependent metrics restricted to cohorts that ship compatible per-voxel labels.
- The full 95-patient BraTS-Africa glioma subset contributes a sub-Saharan African scanner-and-population shift and is the primary leakage-audited external test cohort. BraTS-2023 SSA membership was not an exclusion criterion; all 95 patients were retained.
- UPenn-PDGM contributes an institutional and acquisition-protocol shift. The TCIA release ships in two units: the preprocessed multi-sequence MRI dataset (T1, T1c, T2, FLAIR with brainmask) that we use here, and a separate `automated_segm` subdirectory containing originator-generated labels under a 4-class scheme that we do not harmonise with the BraTS label scheme; UPenn-PDGM therefore enters only via the MRI distribution and serves the unlabelled bottleneck-density, distribution-shift and representation-similarity analyses rather than label-dependent per-voxel ECE, OoD-AUROC or segmentation evaluation.
- MU-Glioma-Post [6] is a post-treatment longitudinal external test cohort and contextual comparator for the radiation-necrosis-like perturbation. It is not a paired

Table S1 Supporting fixed-size sensitivity analysis. Point estimates and patient-level cluster-robust paired-bootstrap 95% confidence intervals for the full external evaluation pools and fixed n=30 sensitivity samples. MU-Glioma-Post observations are longitudinal timepoints clustered by patient.

Cohort	n_{eval}	ΔAURC	95% CI
BraTS-Africa	30 (fixed sensitivity)	−0.022	[−0.038, −0.007]
	95 (full pool)	−0.022	[−0.031, −0.013]
MU-Glioma-Post	30 (fixed sensitivity)	−0.015	[−0.029, −0.002]
	596 (full pool)	−0.015	[−0.018, −0.012]

synthetic–real cohort or a surrogate for the full patient distribution. Longitudinal timepoints are clustered by patient for bootstrap inference.

- Decathlon Task01 [40] contributes 20 simulator parents. Unmodified Decathlon cases are not evaluated as a standalone cohort; only their simulator-derived volumes enter the stress tests.

The reconciled held-out pools comprise 95 BraTS-Africa patients, 509 UPenn-PDGM patients and 596 MU-Glioma-Post longitudinal timepoints from 203 patients. Label-dependent technical readouts are reported only where compatible harmonised BraTS-style per-voxel labels are available. A fixed n=30 external-cohort down-sampling sensitivity replication preserved the directions and three-decimal point estimates of ΔAURC on both external cohorts when compared with their full evaluation pools. Confidence intervals were estimated with patient-level cluster-robust bootstrap, including clustering of repeated MU-Glioma-Post timepoints; no simple timepoint-based square-root scaling was assumed (Table S1).

Leakage protocol.

BraTS-2021 backbone weights, BraTS-2023 GLI training data, and BraTS-Africa (a strict superset of BraTS-2023 SSA) share an upstream collection lineage, so we ran a cross-cohort patient-level dedup audit (perceptual hash plus acquisition-date and ID-prefix matching) over all five cohorts and a separate patient-ID overlap audit against the KAIST BraTS-2021 training-fold patient list [24]. Patient IDs are normalised to the BraTS naming scheme (`BraTS-GLI-XXXXX-YYY` for GLI, `BraTS-SSA-XXXXX-YYY` for Africa, `UPENN-GBM-XXXXX` for UPenn-PDGM, `MUGP-XXXXX-YYY` for MU-Glioma-Post) and prefix-matched against the KAIST BraTS-2021 training fold patient list. All three external cohorts (BraTS-Africa $n_{\text{eval}} = 95$, UPenn-PDGM $n_{\text{eval}} = 509$, MU-Glioma-Post $n_{\text{eval}} = 596$) have 0 patient-ID overlap with the KAIST training fold and 0 strong cross-cohort matches in the dedup audit. Because BraTS-Africa contains the earlier SSA release, SSA membership itself was not an exclusion criterion. The complete 95-patient subset was retained and additionally audited against the 1,251-case BraTS-2023 GLI wrapper-training pool using normalised-prefix matching and perceptual hashing of isotropically resampled brain volumes (Hamming threshold ≤ 5); this comparison found 0 prefix matches. The three external cohorts are therefore leakage-audited against the frozen backbone and wrapper-training pools.

S10 Implementation and statistical plan

All experiments use PyTorch with the AdamW optimiser (learning rate 1×10^{-4}, cosine schedule, weight decay 10^{-5}), 50 epochs per ensemble member, batch size 1, BF16 mixed precision with gradient checkpointing, on 192^3 patches—with a minority at 160^3—at $1\,\mathrm{mm}^3$ isotropic resolution. The augmentation pipeline comprises random affine (rotation $\pm 15°$, scaling $[0.85, 1.15]$, translation ± 5 voxels), random flips along the three principal axes, intensity jitter ($\pm 10\%$), Gaussian noise ($\sigma \leq 0.05$ on z-scored inputs), and gamma correction ($\gamma \in [0.7, 1.5]$). We use publicly available frozen backbone checkpoints: KAIST nnU-Net BraTS-2021 [24] and the SwinUNETR (v1) BraTS-2021 winning weights distributed with MONAI [25]. Frozen-backbone deepest-stage feature tensors are pre-extracted once per cohort/backbone. During head training, the learned $1 \times 1 \times 1$ projection is applied to those fixed tensors to produce $\boldsymbol{f}_v$, so the backbone is not re-run while the projection remains trainable.

The bottleneck dimension is $D{=}256$. The M_1 prototype bank is $P{=}200$ with cosine temperature $\tau{=}10$, calibrated so that training-fold posterior entropy lies in $[0.5, 0.9]$ on average. During cross-validation it is selected using training-fold features only and then frozen for the corresponding validation fold. After selection, τ is fixed for the full-GLI refit; no external or prospective labels are used for temperature selection or retuning. The M_2 ensemble size is $K{=}3$ refinement decoders ($\sim$13.5M trainable parameters each); a SWAG-style low-rank covariance sample [19] is retained from late-training snapshots as an ablation row. The M_3 flow has $L{=}8$ affine-coupling layers, each parameterised by a four-layer MLP of hidden width $d_h{=}512$ with ReLU activation ($\sim$8.2M trainable parameters). Patch sampling during flow training is foreground-balanced at a 60% lesion-overlapping / 40% random-parenchyma ratio. The fusion slope $\rho > 0$ is estimated on an inner GLI calibration subset by minimising Bernoulli NLL for voxel correctness and is then frozen. The reference q_β is the 5th percentile of the applicable GLI ID NLL pool ($\beta = 5\%$), which also supplies μ_{ID}: a training-fold pool is used during cross-validation and the complete GLI pool in the final refit. The MC-Dropout corner uses $T = 8$ stochastic passes. The fixed loss weights are $\lambda_{\mathrm{proto}} = 0.1$ and $\lambda_{\mathrm{flow}} = 0.05$.

Fold-specific wrappers are used only for out-of-fold GLI estimates and cross-validated hyperparameter selection. After all choices are fixed, the mechanism heads, flow, normalisation and NLL references are refitted on all 1,251 GLI cases. Scalar calibration parameters are estimated only on the prespecified inner GLI calibration partition and locked with the resulting wrapper for the Africa, UPenn, MU and prospective evaluations. Thus every non-GLI headline case is scored by one full-GLI-reference model rather than by an unspecified fold combination. The separately fitted per-cohort flows below are used only in a transportability diagnostic. During cross-validation, the 80 GLI and 20 Decathlon simulator parents and all of their derived volumes are assigned wholly to one side of a split. The final full-GLI wrapper has seen the 80 unmodified GLI parent scans but no derived perturbation volume, so its simulator estimate is a perturbation-level rather than parent-level holdout. The synthetic channel-recovery check uses $N \in \{50, 100, 200, 500\}$ and an on-axis-versus-maximum-off-axis Spearman margin of at least 0.05.

Evaluation metrics.

Segmentation quality is reported as whole-tumour Dice and HD95 and is shared across methods because the prediction is inherited from the frozen backbone. Let $\boldsymbol{\ell}_v$ be the frozen backbone's native multiclass logits and let $\mathcal{W}$ index the BraTS whole-tumour subregion classes. The fixed hard whole-tumour prediction is $\widehat{w}_v = \mathbf{1}\{\arg\max_d \ell_{vd} \in \mathcal{W}\}$, and w_v^{ref} is the binary reference union. Voxel correctness is $y_v^{\text{corr}} = \mathbf{1}\{\widehat{w}_v = w_v^{\text{ref}}\}$. For confidence p_v, Brier score is the mean $(p_v - y_v^{\text{corr}})^2$. ECE uses 15 fixed equal-width bins on $[0, 1]$: $\sum_b(|B_b|/N)|\text{acc}(B_b) - \text{conf}(B_b)|$. The operator-selection analysis voxel-pools reference whole-tumour voxels; the headline full-evaluation readout voxel-pools all non-padding voxels inside the brain mask across cases. Foldwise α-selection instead computes ECE (and the Brier sensitivity readout) within each validation patient's non-padding brain mask and then averages patients equally. For the factorial table, full-brain-mask voxel-pooled ECE and Brier are first computed separately in GLI, Africa and MU and the three cohort values are then averaged arithmetically. The calibration-transfer matrix voxel-pools the target evaluation partition's non-padding brain-mask voxels. These are different estimands.

Temperature scaling acts on the frozen backbone's native multiclass logits rather than on an already aggregated scalar confidence. Using the same logits, class set and fixed hard prediction,

$$q_{v,T} = \sum_{c \in \mathcal{W}} \frac{\exp(\ell_{vc}/T)}{\sum_d \exp(\ell_{vd}/T)}, \qquad p_{v,T} = \widehat{w}_v q_{v,T} + (1 - \widehat{w}_v)(1 - q_{v,T}).$$

The mask $\widehat{w}_v$ is never changed by calibration. $T > 0$ is fitted only on an inner GLI training partition and then locked for headline validation and external rows. Because scaling precedes aggregation of multiple tumour classes, $p_{v,T}$ need not preserve the MSP ordering. Source-specific temperatures in the separate calibration-transfer matrix are diagnostic and do not replace this calibrator.

Every comparator in Table 2 is oriented as correctness confidence in $[0, 1]$ for ECE and Brier. MC-Dropout uses c_4^{MCD} from Eq. (S6); SWAG uses the mean posterior-predictive probability assigned to the fixed backbone whole-tumour label; MSP and ODIN use the softmax probability assigned to that label, before and after the ODIN transform, respectively. Energy and ViM scores are oriented so larger values indicate in-distribution support and mapped by a monotone logistic calibrator $\sigma(a+bs)$, $b > 0$, fitted by Bernoulli NLL on the inner GLI partition and then locked. Their AUROC uses the same oriented score, so this monotone mapping does not alter ranking.

Per-voxel OoD-AUROC uses $1 - p_v$ as the OoD score. Reference whole-tumour voxels in the O_1–O_4 derived volumes form the perturbation-positive pool, and reference whole-tumour voxels in each labelled cohort form its negative pool; padding and non-tumour background are excluded. O_5 is excluded from headline AUROC and uses the same reference whole-tumour support when it serves as the pass-through control during α-selection. Point estimates use the voxel pools, whereas uncertainty resampling clusters by simulator parent and cohort patient. Whenever a mechanism map is reduced to one simulator-instance score, that score is the arithmetic mean of the corresponding voxelwise uncertainty signal over reference whole-tumour voxels; every derived instance has non-empty reference whole-tumour support. Selective prediction uses the case confidence defined in Methods, retains cases from high to low confidence and defines case risk as $1 - \mathrm{Dice}_{\mathrm{WT}}$; AURC is the area under this risk–coverage curve.

The supporting technical analyses are estimation-focused. We report effect sizes with 95% confidence intervals from 2,000 patient-level cluster-robust bootstrap resamples [41]; these intervals are not presented as multiplicity-adjusted confirmatory tests.

S11 Supporting retrospective calibration, selective prediction and OoD behaviour

The first quantitative block evaluates calibration, selective prediction and per-voxel OoD detection on the cohorts with compatible BraTS-style labels. BraTS-2023 GLI is reported out of fold as the in-distribution reference; BraTS-Africa and MU-Glioma-Post are labelled external settings evaluated with the locked GLI-reference flow and calibration. The per-cohort-flow comparison reported later is a separate site-adaptation diagnostic and does not alter these headline rows. UPenn-PDGM remains an unlabelled acquisition-protocol sentinel because its preprocessed package lacks compatible harmonised labels for ECE, Brier or OoD-AUROC.

Segmentation quality is inherited from the frozen KAIST BraTS-2021 nnU-Net [24] backbone and is therefore identical to the backbone output: whole-tumour Dice was 0.940 on BraTS-Africa, and whole-tumour HD95 was 1.76 mm on BraTS-2023 GLI, 3.74 mm on BraTS-Africa and 4.49 mm on MU-Glioma-Post. These HD95 values are shared across all trust-map configurations. The clinical triage model does not improve these masks; the supporting analyses below evaluate the trust layer wrapped around them.

Fusion-rule ECE.

On the lesion-restricted GLI validation pool used only to compare aggregation rules for the pure three-mechanism core at $\alpha = 0$, product has the lowest operator-selection ECE, 0.045, compared with 0.052 for mean, 0.061 for maximum and 0.048 for log-pooling (Figure 1). This supports product within that restricted analysis. Its ECE, Brier and AUROC values are not directly compared with the foldwise α-selection, three-cohort factorial or full-pool final-mixture values, which differ in configuration and evaluation scope.

Selective-prediction risk–coverage.

For every method, case confidence is the mean of the lowest-trust decile within the predicted whole-tumour mask, with confidence 0 for an empty mask. Technical analyses retain cases from higher to lower confidence; the clinical risk score is one minus the same final-map aggregation. At coverage c, case risk is $1 - \mathrm{Dice}_{\mathrm{WT}}$ among the retained top-c fraction. AURC is the area under this curve and lower is better. The per-cohort contrast $\Delta\mathrm{AURC} = \mathrm{AURC}_{\mathrm{ours}} - \mathrm{AURC}_{\mathrm{MC\text{-}Dropout}}$ is negative when the final map gives better selective prediction. Table S2 and Figure 2 report paired cluster-bootstrap intervals.

Table S2 Supporting retrospective selective-prediction AURC contrast. For each method, case confidence is the mean of the lowest-trust decile within the predicted whole-tumour mask; an empty mask has confidence 0. Cases are retained from higher to lower confidence and case risk is $1 - \mathrm{Dice}_{\mathrm{WT}}$. The clinical MRI risk is one minus the same final-map aggregation. $\Delta\mathrm{AURC} = \mathrm{AURC}_{\mathrm{ours}} - \mathrm{AURC}_{\mathrm{MC\text{-}Dropout}}$ uses 2,000 patient-clustered paired-bootstrap resamples. All three 95% CIs exclude zero in this retrospective estimation analysis; these results do not replace the prospective clinical endpoint.

Cohort	ΔAURC $\downarrow$	95% CI
BraTS-2023 GLI	−0.018	[−0.032, −0.005]
BraTS-Africa	−0.022	[−0.031, −0.013]
MU-Glioma-Post	−0.015	[−0.018, −0.012]

All three ΔAURC point estimates are negative and all three 95% CIs exclude zero in this retrospective analysis. We report these as estimation results rather than multiplicity-adjusted confirmatory findings. MU-Glioma-Post is the descriptive post-treatment shift read; BraTS-Africa remains the primary leakage-audited external test cohort.

Per-voxel OoD-detection AUROC.

The shared positive instances are the reference whole-tumour voxels from derived O_1–O_4 volumes; padding and non-tumour background are excluded, as is the unmodified O_5 pass-through control. Each labelled cohort supplies its own reference-negative whole-tumour voxel pool, so the three rows use different negative references rather than three simulators. AUROC uses $1 -$ confidence as the OoD score. Voxel pools give point estimates, while interval resampling is clustered by simulator parent and cohort patient. The final map outperforms MC-Dropout on all three labelled cohorts (Table 2), with mean delta +0.052 (GLI +0.045, Africa +0.057, MU +0.054). UPenn-PDGM lacks the companion labels. The perturbation-pool AUROC floor is 0.65 on the primary cohort; the final map reaches 0.721 on BraTS-Africa, 0.071 above that floor.

S12 MC-Dropout corner-selection witness

The information-set bound in Proposition S1 does not guarantee performance of the scalar α family, so two GLI-only empirical questions are evaluated without consuming external-test data: (i) whether held-out selection chooses an interior coefficient rather than the $\alpha = 1$ MC-Dropout corner, and (ii) whether the selected mixture improves the observed validation objective and headline retrospective estimates. Mechanism-specific ablations are interpreted separately and do not identify which channel supplies any gain over MC-Dropout.

Five-fold patient-level cross-validation over the full 1,251-case GLI pool selects $\alpha^{\star} = 0.6$ in all folds (Table S3). The mean foldwise val-ECE gap versus $\alpha = 1$ is

Table S3 Five-fold val-pool $\alpha^\star$ selection. Patient-level GLI cross-validation uses approximately 1,000 training and 250 validation patients per fold; simulator parents and all of their derived volumes remain wholly within one side of a split. The grid is $\{0, 0.2, 0.4, 0.6, 0.8, 1.0\}$ and all folds select $\alpha^\star = 0.6$. The mean val-ECE 0.160 and gap 0.106 ± 0.007 use per-patient brain-mask ECE followed by equal patient averaging and are specific to this foldwise mixture-selection estimand and are not directly compared with lesion-restricted operator-selection, factorial-core or full-pool headline ECE. A Brier selection sensitivity check reproduces $\alpha^\star = 0.6$ in all folds.

Fold	$\alpha^\star$	val-ECE@$\alpha^\star$	val-ECE@$\alpha = 1$	gap
1	0.6	0.166	0.263	+0.097
2	0.6	0.162	0.268	+0.106
3	0.6	0.161	0.264	+0.103
4	0.6	0.158	0.268	+0.110
5	0.6	0.155	0.269	+0.114
mean	0.6	0.160	0.266	$+0.106 \pm 0.007$

0.106 ± 0.007, about a 40% relative reduction. This patient-equal, brain-mask val-ECE of 0.160 belongs to the foldwise α-selection estimand and is not directly comparable with the lesion-restricted core ECE 0.045, factorial-core mean 0.370 or headline final-mixture ECE 0.385. Val-AUROC is tied across α in this sweep. External transfer is evaluated directly with the locked operating point.

The attribution analysis below compares only c_1, c_2, c_3 and therefore cannot establish whether any core channel is a re-expression of c_4^{MCD}. It shows heterogeneous dominance within the core, while the final-mixture-versus-MC-Dropout rows in Table 2 provide the aggregate retrospective comparison. Neither analysis establishes probabilistic independence or attributes the mixture-level difference to one channel.

S13 Mechanism-resolved stress-test evidence

The mechanism-resolved analyses examine the three-mechanism core, separately from the four-channel final map. First, core mechanisms are tested for descriptive non-redundancy. Second, perturbation conditions are examined for dominant or directional channels, motivating mechanism-channel-specific reader and workflow studies.

Empirical mechanism distinctness.

A seven-cell $\alpha = 0$ factorial ablation examines strict subsets of M_1, M_2 and M_3 without MC-Dropout (Table S4). Each leave-one-out point estimate loses at least 0.01 in OoD-AUROC relative to the full core; the largest drop is on M_2 removal (-0.056) and the smallest on M_1 removal (-0.023). These descriptive estimates support operationally non-identical contributions in the tested setting but do not establish conditional independence or causal necessity.

Table S4 Factorial three-mechanism-core ablation. All rows are $\alpha = 0$ core configurations without MC-Dropout mixing and report arithmetic means across GLI, BraTS-Africa and MU-Glioma-Post. ECE and Brier use each cohort's non-padding brain mask; OoD-AUROC compares derived O_1–O_4 whole-tumour voxels with the corresponding cohort reference voxels. ECE and Brier were not evaluated for the M_1-only cell.

Mechanism cell	**Cohort**	**ECE ↓**	**Brier ↓**	**OoD-AUROC ↑**
M_1 only	3-cohort mean	—	—	0.612
M_2 only	3-cohort mean	0.3878	0.1933	0.681
M_3 only	3-cohort mean	0.3812	0.1937	0.594
M_1+M_2	3-cohort mean	0.3779	0.1916	0.683
M_1+M_3	3-cohort mean	0.3744	0.1908	0.668
M_2+M_3	3-cohort mean	0.3768	0.1912	0.701
$M_1+M_2+M_3$	3-cohort mean	0.3700	0.1880	0.724

Table S5 Per-perturbation dominant-mechanism breakdown. Fractions are computed over derived instances with 95% parent-clustered paired-bootstrap CIs. O_1 is dominated by M_2, O_2 by M_1, and O_3 shows directional M_3 low-density-support dominance with overlapping intervals.

Perturbation condition	$f_{M_1\text{-dom}}$ (%)	$f_{M_2\text{-dom}}$ (%)	$f_{M_3\text{-dom}}$ (%)
O_1 rad-necr.	17 $[8, 27]$	80 $[70, 90]$	3 $[0, 8]$
O_2 abscess	65 $[53, 77]$	35 $[23, 47]$	0 $[0, 0]$
O_3 PCNSL	32 $[20, 43]$	25 $[15, 37]$	43 $[32, 57]$

Per-perturbation dominant mechanism.

We apply M_1, M_2 and M_3 independently to O_1–O_3 ($n_{\text{instances}} = 60$ per condition), sampled with parent stratification from 300 derived volume instances per condition. For each derived instance, the M_1, M_2 or M_3 score is the arithmetic mean of that mechanism's voxelwise uncertainty signal over reference whole-tumour voxels. We z-score these instance scores across the pooled subset and define the dominant mechanism as the argmax. All intervals resample parent clusters, so derived instances are not treated as independent patients. The CIs of the dominant and non-dominant mechanisms do not overlap on O_1 (M_2 80% $[70, 90]$ vs M_1 17% $[8, 27]$, Δ_{frac}=63 pp) and on O_2 (M_1 65% $[53, 77]$ vs M_2 35% $[23, 47]$, Δ_{frac}=30 pp, with CIs only marginally non-overlapping at 95%); on O_3 the dominant M_3 43% $[32, 57]$ has overlapping CIs with the non-dominant M_1 32% $[20, 43]$ at the lower end, so we report the O_3 specialisation as a directional finding rather than a strict-CI claim. The pattern is directionally consistent with the intended observables: radiation-necrosis-like perturbations may increase ensemble disagreement, abscess-like intensity signatures may depart from prototype geometry, and PCNSL-like features may occupy a low-density region. These explanations are post hoc interpretations of instance-level argmax patterns, not causal tests. The complete breakdown is reported in Table S5.

Synthetic channel-recovery check.

Across $N \in \{50, 100, 200, 500\}$, M_1 correlates with the planted prototype-geometry direction a_{proto} at $\rho \geq 0.77$, and M_3 with the planted low-density direction a_{flow} at $\rho \geq 0.79$. The on-axis-versus-maximum-off-axis margin is at least 0.05 at all sample sizes, but this is a synthetic recovery check rather than identification of causal uncertainty sources.

The corresponding qualitative spatial decomposition is presented in main-text Figure 3; the quantitative mechanism analyses above use the prespecified evaluation pools.

Interpretation of the supporting mechanism analyses.

Two perturbation conditions show clear dominant core channels and a third shows directional low-density-channel dominance. This factorisable core exposes per-channel attributions unavailable from the single-channel comparators. Such attributions were not displayed or separately tested in the present reader study; their value requires a future mechanism-channel-specific reader and workflow evaluation.

S14 MIMIC-OOD-3D-Brain simulator-derived stress-test set

A simulator-derived stress-test set is useful only if its perturbations are not mistaken for patient-diagnosed mimic cohorts. We use two validation checks. F_1 is a perturbation-selectivity check asking whether the locked-GLI S^{flow} score separates the unmodified O_5 pipeline control from O_1–O_3. F_2 is a descriptor-plausibility gate: for each morphology-bearing condition, at least three of four intensity descriptors must have Bhattacharyya coefficient ≥ 0.50 and all four shape descriptors must reach ≥ 0.70. F_2 supports descriptor overlap only, not paired synthetic–real uncertainty correspondence or patient-distribution surrogacy.

The set comprises 1,500 derived NIfTI volume instances from 100 parent cases: 80 GLI and 20 Decathlon parents. Conditions O_1–O_4 each contribute 300 volumes (100 parents times three severity levels); O_5 contributes 300 pass-through volumes (three matched replicates per parent, with no severity meaning). Thus 1,500 counts derived instances, not independent patients. Simulator materials are available from the corresponding author on reasonable request, subject to source licences and data-use agreements. Because the materials are request-only, we refer to a simulator-derived stress-test set rather than a public benchmark. The perturbations do not replace real abscess, PCNSL, metastasis, demyelination, infarct or treatment-related-change cohorts. The simulated conditions and descriptor checks are summarised in Figure S1.

Figure S1(b)–(c) compare $n = 60$ real GLI cases with $n = 300$ derived instances per morphology-bearing condition. O_1–O_3 each reach Bhattacharyya coefficient ≥ 0.50 on at least three of four intensity descriptors and ≥ 0.70 on all four shape descriptors, so all three pass F_2. This supports descriptor-level plausibility only; it does not establish a paired synthetic–real uncertainty association or patient-distribution surrogacy. Prospectively assembled real-mimic cohorts remain the target for clinical validation.

F_1 perturbation-selectivity check.

Using the locked GLI-reference S^{flow} map, the predefined check assigns each derived volume the arithmetic mean of S_v^{flow} over its reference whole-tumour voxels and compares 60 O_5 pass-through instances with 180 O_1–O_3 instances. AUROC is 0.773, above the 0.55 floor. Resampling is clustered by parent. This is a planted-perturbation selectivity check, not a strict ID-versus-OoD cohort test; O_5 remains excluded from the headline OoD-AUROC analyses.

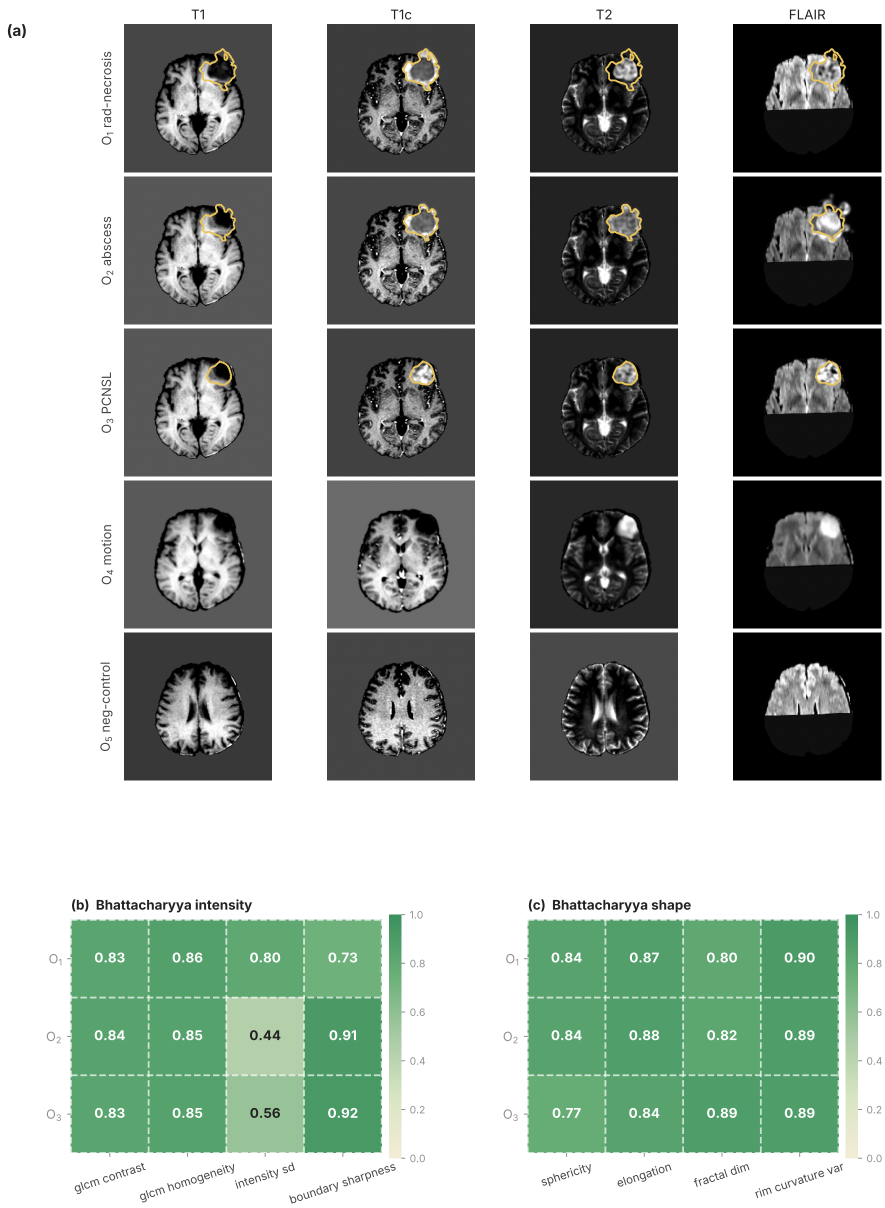

Fig. S1 MIMIC-OOD-3D-Brain simulator and descriptor checks. (a) Example slices for four perturbation conditions (O_1 radiation-necrosis-like, O_2 abscess-like, O_3 PCNSL-like and O_4 motion artefact) plus the unmodified O_5 pipeline control, in four MRI sequences. For O_1–O_3 only, the orange contour marks the lesion region receiving a modification; O_4 is a whole-image acquisition perturbation and O_5 is unmodified. **(b)** Heatmap of Bhattacharyya coefficient on intensity descriptors between simulated and real BraTS-2023 GLI lesion descriptors for the three morphology-bearing mimic classes (O_1–O_3). **(c)** Same heatmap for shape descriptors. Predefined acceptance: intensity descriptors must reach ≥ 0.50 on $\geq 3/4$ descriptors AND shape descriptors must reach ≥ 0.70 on all 4 morphology-discriminating descriptors. All three classes satisfy these criteria.

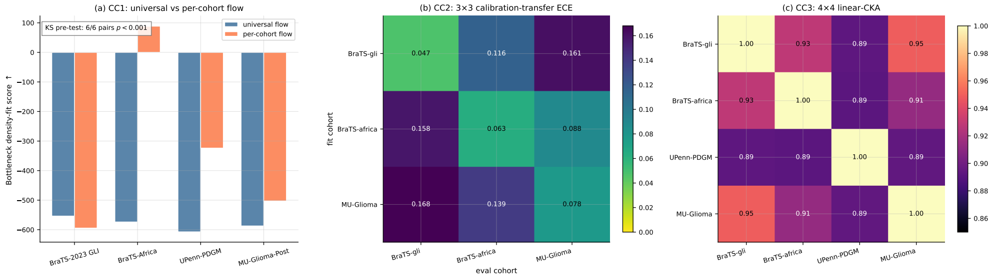

Fig. S2 Three supporting cross-cohort transportability diagnostics for the imaging trust pipeline. (a) Held-out negative-mean-NLL density-fit scores for a pooled universal flow and separately fitted per-cohort flows; higher (less negative) is better. This diagnostic does not replace the locked GLI-reference flow. A descriptive KS check on each patient's mean locked-reference NLL rejects equality for all six cohort pairs at $p < 0.001$. **(b)** Temperature-only calibration-transfer ECE. Each row calibrator is fitted on a source-cohort fit partition and evaluated on a disjoint column-cohort partition; diagonal cells are also fit/evaluation separated. This diagnostic is not directly comparable with any final-mixture ECE and does not retune headline rows. **(c)** Channel-aligned linear CKA across the four glioma cohorts, using the shared 256 locked projected-bottleneck channels as aligned observations as defined in Methods. PICTURE has a different channel basis and is not quantitatively compared. Higher values indicate greater similarity.

S15 Cross-cohort transferability

The main external-cohort evaluation uses the GLI-reference flow, normalisation and calibration unchanged on BraTS-Africa and MU-Glioma-Post. The analyses below are three separate supporting transportability diagnostics for the imaging trust pipeline: descriptive density adaptation, temperature-calibration transfer and channel-aligned representation similarity. Per-cohort fits in these diagnostics do not enter the headline external trust maps. PICTURE pathology features [37] have a different channel basis and dimension; they are not an input to patient-level fusion and are not quantitatively compared with the MRI CKA matrix. The three diagnostics are summarised in Figure S2.

Universal-versus-per-cohort flow comparison.

This separate diagnostic compares a flow fitted to pooled four-cohort features with cohort-specific flows. All density-fit scores are negative mean NLL in natural-log nats per 256-dimensional voxel feature, averaged over patient-disjoint held-out feature vectors; higher, less-negative values indicate better descriptive fit, and each reported Δ is per-cohort minus universal. Longitudinal observations remain grouped by patient.

The accompanying KS diagnostic is one-dimensional: for each patient we average locked-reference-flow NLL within the brain mask and compare this scalar across the six cohort pairs. All six descriptive comparisons have $p < 0.001$. This does not constitute a multivariate KS test on 256-D features or a confirmatory decision rule.

Cohort-specific flows have higher held-out density-fit scores on three of four cohorts (BraTS-Africa $\Delta = 663$, UPenn-PDGM $\Delta = 283$, MU-Glioma-Post $\Delta = 84$); the pooled flow is higher on GLI ($\Delta = -41$), which contributes the largest share of the pooled reference. These results motivate prospective evaluation of site-adapted density fitting but do not replace the locked GLI-reference flow in the main external analysis.

Calibration-transfer matrix.

The 3×3 matrix spans GLI, BraTS-Africa and MU-Glioma-Post and asks how a temperature fitted on source cohort A transfers to target cohort B. Each cell voxel-pools the disjoint target evaluation partition's non-padding brain mask against whole-tumour voxel correctness. Within-cohort diagonal ECE ranges from 0.047 to 0.078, while off-diagonal ECE ranges from 0.088 to 0.168. For each transfer, degradation is defined against the target diagonal:

$$R_{A \to B} = \frac{\mathrm{ECE}_{A \to B}}{\mathrm{ECE}_{B \to B}}.$$

The six off-diagonal ratios range from $1.1\times$ to $3.6\times$ with median approximately $2.1\times$. These temperature-only, fit/evaluation-separated diagnostic ECEs are not directly comparable with operator-selection, foldwise mixture-selection, factorial-core or headline final-mixture ECE. They motivate local prospective calibration assessment rather than retroactive retuning of the reported rows.

Bottleneck-similarity matrix.

The channel-aligned CKA defined in Methods compares inter-channel Gram structure without requiring paired patients or equal cohort sizes. The four glioma cohorts have pairwise similarities 0.89–0.95. This descriptive similarity does not itself guarantee transfer of density or calibration. PICTURE is not inserted into the matrix because its 768-dimensional representation has neither the same channel count nor a shared channel basis with the 256-channel MRI backbone. Overall, cohort-specific flows have the higher descriptive fit on three of four cohorts, the within-disease CKA values are 0.89–0.95, and target-diagonal calibration-degradation ratios are $1.1\times$–$3.6\times$ with median approximately $2.1\times$. These separate diagnostics support prospective local assessment before deployment; they do not modify the locked external results.

S16 Auxiliary analyses

Fusion-operator ablation.

We report ECE, Brier and OoD-AUROC for product, mean, maximum and log-pooling on the GLI cross-fold validation pool restricted to reference whole-tumour voxels. This is a pure three-mechanism-core analysis at $\alpha = 0$; its metrics are not directly compared with the foldwise mixture-selection, factorial-core means or full-pool headline mixture. Dice is invariant because segmentation is inherited from the frozen backbone. The product operator achieves the lowest ECE and highest OoD AUROC: ECE 0.045 (vs mean 0.052, max 0.061, log-pool 0.048); Brier 0.083 (vs 0.092, 0.102, 0.087); and OoD AUROC 0.840 (vs 0.810, 0.790, 0.830). Max-pooling underperforms on all three trust metrics, consistent with its lossy single-factor reduction; mean fusion is competitive on calibration but trails on OoD AUROC by 0.030; log-pooling sits between mean and product on both metrics. The product operator performed best for calibration and ranking in this empirical validation; this result does not by itself establish conditional independence among mechanism channels.

Table S6 Trust-map computational footprint. Trainable parameters, peak training VRAM and per-patch forward latency of the trust-map heads on top of the frozen backbone, under BF16 mixed precision and 192^3 patches. The table does not report total training time. A dash denotes no separately trainable parameters or no separately reported incremental peak allocation; MC-Dropout sampling latency is included in the pipeline total.

Component	Params	VRAM (GB)	Forward (s)
Frozen backbone [24]	—	11.2	0.18
M_1 prototype-entropy head (P=200)	∼0.1 M	0.4	0.02
M_2 deep ensemble (K=3)	∼40.5 M	7.6	0.18
M_3 RealNVP density head	∼8.2 M	1.4	0.04
MC-Dropout (T=8 samples)	—	—	0.08
Pipeline total	∼48.8 M	20.6	∼0.5

Compute footprint.

Table S6 reports trainable parameters, peak training VRAM and per-patch forward latency by component; it does not report total training time. The backbone is frozen and features are pre-extracted, so the table separates the fixed backbone footprint from trainable uncertainty heads.

Bottleneck-feature t-SNE across glioma cohorts.

As a descriptive complement to the CKA bottleneck-similarity panel of Figure S2(c), Figure S3 projects the bottleneck features of the four glioma cohorts to two dimensions with t-SNE. The four glioma cohorts intermix at the colour-mixing level within several clusters of the projected space, consistent with the within-disease cluster reported by CKA at pairwise similarity 0.89–0.95. The PICTURE pathology bottleneck [37] is not co-projected or quantitatively compared because of the 768-d versus 256-d dimensionality and channel-basis mismatch noted in the Cross-cohort transferability section.

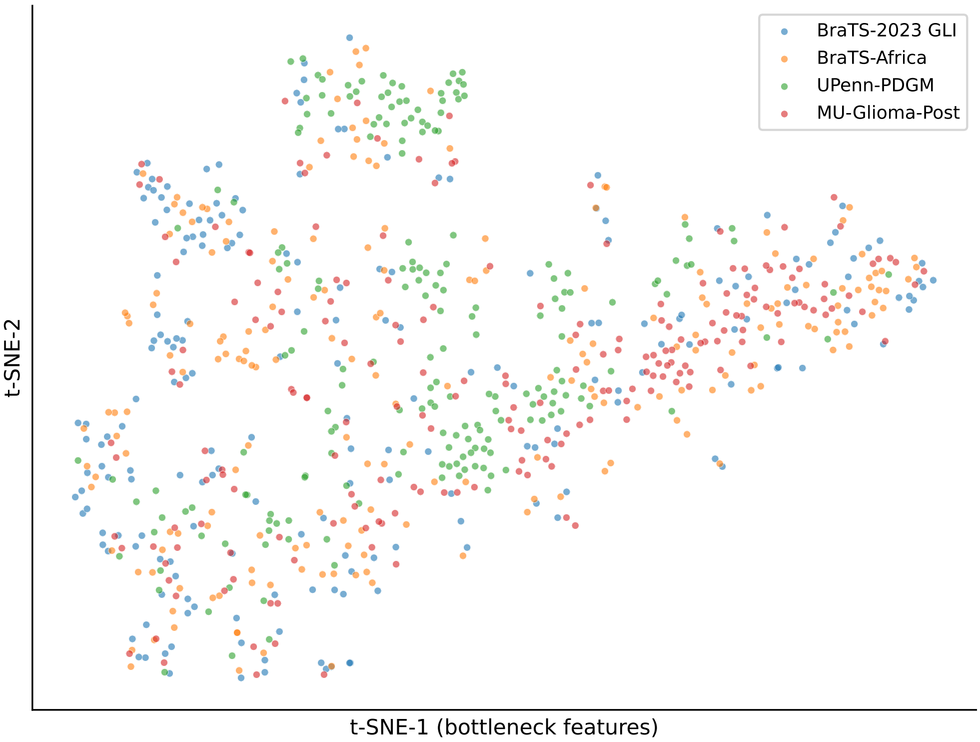

Fig. S3 Bottleneck-feature t-SNE across glioma cohorts. t-SNE projection of the bottleneck features of the four glioma cohorts (BraTS-2023 GLI, BraTS-Africa, UPenn-PDGM, MU-Glioma-Post). The four cohorts intermix at the colour-mixing level within the projected clusters, consistent with the within-disease CKA cluster of Figure S2(c). The PICTURE pathology bottleneck is not overlaid because of the 768-d vs 256-d dimensionality mismatch.

Backbone validity vs concurrent SOTA segmenters.

Because we use the frozen KAIST BraTS-2021 winner as the segmentation backbone [24], our whole-tumour Dice 0.940 on BraTS-Africa is at the upper end of the published BraTS-2021 / BraTS-2023 Dice ranges for nnFormer (∼0.91 [42]), SegMamba (∼0.92 [43]), U-Mamba (∼0.91 [44]), and SwinUNETR-V2 (∼0.91 [45]). These comparator numbers are cited from each paper's BraTS-2021 / BraTS-2023 evaluation rather than recomputed on our cohort splits, so the cohort set differs; an apples-to-apples re-evaluation on BraTS-Africa can be performed with the benchmark evaluation harness available from the corresponding author upon reasonable request. As a like-for-like reference point we additionally run the public BraTS-2021 winning SwinUNETR-V1 5-fold ensemble [45] on the same full leakage-audited BraTS-Africa evaluation cohort using its publicly released weights; that ensemble reaches WT-Dice 0.856 on the same split (Table S7), confirming the KAIST 0.940 figure is at the high end of the publicly-reproducible same-split range. This same-split comparison supports the choice of a strong frozen backbone; it is a technical validity check and not evidence that the review-triage wrapper improves segmentation accuracy.

Table S7 Backbone Dice vs concurrent SwinUNETR-V1 ensemble. Whole-tumour Dice on the full leakage-audited evaluation cohorts (BraTS-Africa $n = 95$, MU-Glioma-Post $n = 596$) plus the BraTS-2023 GLI out-of-fold reference pool ($n = 1{,}251$): the KAIST 1st-place BraTS-2021 backbone used in this work versus a same-split run of the public SwinUNETR-V1 5-fold ensemble.

Cohort	KAIST 1st-place	SwinUNETR-V1
BraTS-Africa	0.940	0.856
BraTS-2023 GLI	0.934	0.868
MU-Glioma-Post	0.857	0.859

Selection-objective, ensemble-size and val→test robustness.

Three sensitivity checks evaluate the reported technical operating point. (i) Selection objective. Rerunning the 5-fold sweep of Table S3 with $\alpha^\star$ chosen by the Gneiting–Raftery proper-scoring-rule objective $J_{\text{proper}}(\alpha) = -\text{val}-\text{Brier}(\alpha)$ instead of the heuristic ECE$-$AUROC objective yields the identical fold-by-fold assignment $\alpha^\star_{\text{Brier}}$=0.6 on all 5/5 folds, with a cross-fold mean val-Brier gap of 0.048 ± 0.003 (val-Brier 0.295 at $\alpha^\star$ vs 0.343 at the MC-Dropout corner). The $\alpha^\star$=0.6 choice is therefore robust to the selection-objective family. (ii) Ensemble size. On the fixed 30-case BraTS-Africa sensitivity slice with a 240-instance perturbation pool, $K = 5$ (seeds $\{42, 1337, 2026, 11, 7\}$) reproduces the $K = 3$ OoD-AUROC 0.721 to three decimals (ΔAUROC = 0.000). This limited comparison over $K \in \{3, 5\}$ does not establish ensemble-size saturation or determine whether the other mechanisms remain necessary for larger ensembles. (iii) Val→test transfer. The selected $\alpha^\star = 0.6$, product core operator, GLI-reference flow, normalisation and headline calibrator are applied unchanged to BraTS-Africa and MU-Glioma-Post; UPenn-PDGM remains an unlabelled shift diagnostic. The separate per-cohort-flow analysis is a site-adaptation diagnostic and does not feed these rows. With the locked operating point, all three 95% CIs for ΔAURC exclude zero in the retrospective estimation analysis.

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